Auburn Coherence Framework v1.2

March 2026 Prediction Verification

Lookback Analysis: Seven-Region Global Assessment

With Cumulative February–March Reclassification

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This document compares geometric constraint predictions issued by the Auburn Coherence Framework on February 5, 2026 against observed meteorological events through March 25, 2026. It constitutes the second monthly verification report in a planned series covering the February–May 2026 prediction window. All observed data are sourced from national meteorological agencies, intergovernmental bodies, and peer-reviewed monitoring platforms. This report also provides a cumulative reclassification of the original 28 predictions assessed in the February 2026 verification report, incorporating two months of observational data.

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1 Executive Summary

On February 5, 2026, the Auburn Coherence Framework v1.2 issued geometric constraint predictions for atmospheric dynamics across seven global regions, covering a forward window of February 5 through May 5, 2026. The February 2026 verification report, published in March 2026, assessed 28 predictions whose primary windows intersected February. This document extends that assessment through March 25, 2026, evaluating both (a) predictions whose windows opened or continued through March, and (b) the cumulative reclassification of the original 28 predictions using two months of observational data.

1.1 March Standalone Results

Of the 15 predictions whose windows intersected March and produced decisive new verification data:

Table 1: March 2026 Standalone Verification Summary

Verification Status	Count	Percentage
Confirmed	8	53.3%
Confirmed (Exceeded)	2	13.3%
Partial	3	20.0%
Developing (window open)	1	6.7%
Not Verified	1	6.7%
Total assessed	**15**	**100%**

Ten of 15 predictions were confirmed or confirmed with exceedance (66.7%). Three exhibited partial verification with correct mechanism identification but geographic or magnitude deviation (20.0%). One prediction remains within an open window with confirmed precursor signals (6.7%). One prediction was not verified (6.7%)—the only clean miss across two months of assessment.

1.2 Cumulative Reclassification (February–March)

Applying two months of observational data to the original 28 February-assessed predictions:

Table 2: Cumulative February–March Reclassification (Original 28 Predictions)

Status	Feb (Published)	Cumulative	Change
Confirmed + Confirmed Early	20 (71.4%)	21 (75.0%)	+1
Partial	5 (17.9%)	4 (14.3%)	−1
Developing	3 (10.7%)	3 (10.7%)	0
Not Verified	0 (0.0%)	0 (0.0%)	0

The cumulative confirmed rate rose from 71.4% to 75.0% on the same prediction set. The increase reflects AU-1 (Queensland tropical cyclone) upgrading from Partial to Confirmed following Tropical Cyclone Narelle's Category 4 Queensland landfall on March 20—within the original prediction window of February 15–April 5. No prediction confirmed in February was contradicted by March observations.

1.3 Key March Events

Five predictions confirmed in February escalated in severity during March rather than regressing:

- **AU-1:** Tropical Cyclone Narelle made a rare triple Australian landfall (QLD Cat 4, NT Cat 3, WA Cat 3–4 in progress), only the 4th such event in Australian history. Peak intensity reached Category 5.
- **US-5/US-7:** Western U.S. snowpack reached all-time SNOTEL-era lows across multiple basins. A concurrent historic heat dome shattered March temperature records in 14 states. Separately, the March 5–10 tornado outbreaks produced an EF5 in Indiana—the first since 2013.
- **SA-2:** Peru's Coastal El Niño emergency expanded from 283 to 1,177 districts in state of emergency (62% of all districts nationwide). Ecuador declared a national state of emergency.
- **ME-2:** A "virtually unheard of" storm system struck the Middle East March 25–31, with models projecting up to 500 mm across Iraq—potentially a year's rainfall in three days.

1.4 Aggregate Impact

Aggregate Impact

Events covered by framework predictions during March 2026 were associated with an estimated USD $8–12 billion in combined economic losses and more than 120 additional fatalities across seven regions. The largest single-region losses included the Western U.S. heat dome and wildfire complex (Nebraska's largest fire in state history: 643,074 acres), Peru's escalating El Niño Costero emergency (92 dead, 196,483 affected), and Tropical Cyclone Narelle's triple landfall threatening Pilbara LNG and iron ore infrastructure. Combined with February's USD $12–15 billion estimate, the two-month cumulative impact exceeds USD $20–25 billion across the seven assessed regions.

2 Methodology

The methodology employed in this report is identical to the February 2026 verification report. This section summarizes the framework for readers encountering the series for the first time.

2.1 Prediction Generation

Predictions were generated using the Auburn Coherence Framework v1.2, which applies geometric constraint analysis to atmospheric dynamics. The framework computes coherence values $C(t)$ representing the degree of geometric alignment within identified atmospheric structures. Each prediction was issued with a probability estimate, coherence value, temporal window, geographic scope, and mechanism description. The governing equations, coherence decay rates (λ_d), and geometric alignment tensor specifications remain proprietary.

2.2 Verification Standard

Observed events were compiled from official national meteorological agencies, intergovernmental monitoring bodies, and peer-reviewed disaster tracking platforms. Primary sources for March 2026 include:

- **Australia:** Bureau of Meteorology (BOM), SecureNT, Climate Council, ABARES

- **China:** China Meteorological Administration (CMA), National Meteorological Centre (NMC)
- **United States:** National Weather Service (NWS), NOAA CPC, Storm Prediction Center (SPC), NRCS SNOTEL, Drought.gov, World Weather Attribution (WWA)
- **Western Europe:** AEMET (Spain), IPMA (Portugal), Met Office (UK), EAWS, Environment Agency (England), Copernicus CAMS, NASA Earth Observatory
- **Middle East:** GDACS/ECHO, Muscat Daily, CDAA (Oman), NCM (UAE)
- **South America:** INMET (Brazil), INDECI (Peru), ENFEN, OCHA, Buenos Aires Grain Exchange
- **Southeast Asia:** PAGASA, JMA, JTWC, BMKG (Indonesia), BNPB, AHA Centre, TMD (Thailand)
- **Global:** IRI/Columbia, NOAA CPC, Copernicus C3S, BOM ENSO Outlook

2.3 Classification Definitions

The same five-tier classification scheme from the February report is applied:

Table 3: Verification Classification Definitions

Status	Definition
Confirmed	Event occurred within the specified window, region, and mechanism. Observed magnitude consistent with or exceeding the predicted impact class.
Confirmed (Early)	Event occurred before the specified window opened but matched the predicted mechanism, region, and impact class.
Confirmed (Exceeded)	Event confirmed within window with observed magnitude substantially exceeding the predicted threshold. New classification introduced for March to distinguish escalated outcomes.
Partial	The predicted mechanism was observed, but the event exhibited meaningful geographic displacement, reduced magnitude, or altered hazard type.
Developing	The prediction window remains open. Precursor signals confirmed but primary event has not yet occurred.
Not Verified	No observed event matched the predicted mechanism, region, or impact class within the specified window.

Honest Framing

The addition of "Confirmed (Exceeded)" as a classification is a methodological refinement for the March report. It is applied when observed outcomes substantially surpass the predicted threshold—for example, when a prediction of <75% snowpack verifies at all-time record lows across multiple basins, or when a Cat 3+ cyclone prediction verifies as Cat 4–5 with a historically rare triple landfall. This classification is counted identically to "Confirmed" in all aggregate statistics. Its purpose is descriptive transparency, not statistical inflation.

3 Global Precursor State — March 2026 Update

The February verification report confirmed six of seven global precursor parameters. This section updates each parameter through March 25, 2026, documenting the evolution of boundary conditions that govern downstream regional predictions.

3.1 ENSO State

Table 4: ENSO Verification — March Update

Parameter	Feb Status	Mar Observed	Update
ONI	Weak La Niña (≤ -0.5°C)	Niño 3.4 at -0.0°C (week of Mar 11)	Neutral achieved
RONI (new metric)	N/A	DJF -0.90°C	NOAA adopted Feb 1
Transition timing	Neutral by Feb–Apr	IRI: 91% neutral MAM	Confirmed
Subsurface Kelvin wave	Propagating	Niño 1+2 at +1.5°C	Intensifying
El Niño emergence	Mid-2026	IRI: 62% by JJA 2026; 72–80% by SON	On track
Strong El Niño	Not assessed	1-in-3 chance ($\geq +1.5$°C) by OND 2026	New signal

The ENSO transition proceeded as the framework anticipated. The Niño 3.4 index reached -0.0°C for the week centered March 11, confirming the effective end of oceanic La Niña conditions. NOAA adopted the Relative Oceanic Niño Index (RONI) on February 1 as its primary tracking metric, replacing the traditional ONI after 75 years; under RONI, the DJF 2025–26 value was -0.90°C (stronger than ONI's -0.45°C for the same period). BOM stated the 2025–26 La Niña was "close to its end" with neutral expected in early austral autumn.

The transition toward El Niño is building faster than most models projected at issuance. A massive subsurface warm pool has developed in the western equatorial Pacific at 100–250 m depth, with eastward-propagating Kelvin waves eroding the cold anomaly. The Niño 1+2 index at +1.5°C reflects this warming reaching the eastern Pacific surface—a signal the framework identified on February 5 as the "most significant El Niño precursor since the April–June 2023 buildup." CPC/IRI consensus forecasts now assign 62% probability of El Niño by JJA 2026 and 72–80% by SON 2026, with a 1-in-3 chance of a strong event ($\geq +1.5$°C) by OND 2026.

3.2 Stratospheric State — Second SSW Event

Table 5: Stratospheric State — March Update

Parameter	Feb Status	Mar Observed	Update
SSW occurrence	Confirmed (Jan–Feb)	Second major SSW early Mar	Confirmed
Vortex config.	Split confirmed	Split into two lobes by Mar 4	Intensified
10 hPa temps	∼40°C above normal	Surge from ∼205 K to ∼230 K	Confirmed
Zonal wind	Confirmed	Westerly to easterly at 60°N	SSW criteria met
Surface impacts	Blizzard of 2026 (Feb)	Mar blizzard (3–4 ft); late-Mar European cold plunge	Continued
AO response	Negative	Strat. AO strongly neg.; surface AO pos. mid-Mar	Decoupled

A second major Sudden Stratospheric Warming event was confirmed in early March, with 10 hPa Arctic temperatures surging approximately 25 K and zonal winds reversing from westerly to easterly at 60°N—meeting the formal definition of a major SSW. The polar vortex split into two lobes by March 4. Rather than fully dissipating, the SSW evolved toward what may be reclassified as the Final Warming (seasonal polar vortex breakdown).

Surface impacts were substantial and geographically distributed. The March 13–17 U.S. blizzard—with 3–4 feet of snow across the Upper Midwest—was directly linked to the SSW-driven pattern. By late March, AER's Dr. Judah Cohen described a developing "Great March Reversal" in which Greenland blocking would channel Arctic air into Western and Central Europe, producing 15–20°C temperature drops within two days and threatening vineyards and orchards that had already begun budding. The stratospheric AO remained strongly negative while the surface AO was briefly positive at mid-month, indicating a decoupled state that typically resolves toward surface negative AO within 1–2 weeks.

3.3 Global Sea Surface Temperatures

Table 6: Global SST Verification — March Update

Parameter	Observed (Feb–Mar 2026)	Status
Global SST anomaly	Feb 2026: 20.88°C (+0.41°C); joint 2nd warmest Feb	Confirmed
Global mean temperature	Feb 2026: 13.26°C (+1.49°C above pre-industrial); 5th warmest Feb	Confirmed
North Atlantic	3rd longest marine heatwave for January on record	Confirmed
NE Pacific MHW	Largest MHW by area since 1982 monitoring (∼10M km^2)	New signal
Coral Sea SST	All-time record 29.5°C (Feb); fueled TC Narelle	Confirmed
Peru coastal SST	Niño 1+2 at +1.5°C (Mar 11)	Intensifying
Global MHW coverage	23% of ocean in Jan; projected 37% by year-end	Confirmed

Global ocean temperatures remained at historically elevated levels. The Copernicus C3S ERA5 dataset recorded February 2026 as the 5th warmest on record globally (+1.49°C above pre-industrial) with SSTs (60°S–60°N) tying February 2025 as the joint 2nd warmest February. The 12-month running average stood at +1.44°C above pre-industrial.

The Northeast Pacific marine heatwave (NEP25A), which emerged in May 2025, remained the largest MHW by area observed since satellite monitoring began in 1982, covering approximately 10 million km^2 (NOAA SWFSC). This persistent warm anomaly contributed to the atmospheric ridge pattern that produced the March heat dome over the Western United States.

The Coral Sea SST record of 29.5°C set in February provided the thermodynamic fuel for Tropical Cyclone Narelle's rapid intensification to Category 5—a direct link between the framework's global SST assessment and the regional cyclone prediction.

3.4 MJO State

The MJO was active through March, progressing through Phase 8 (Western Hemisphere/Africa) by mid-March and transitioning into the Western Pacific by March 22 at moderate strength. Strong low-level westerly winds associated with MJO/Rossby wave interaction across the equatorial West Pacific may further support the El Niño transition through additional Kelvin wave generation. The MJO's March trajectory was broadly consistent with the framework's February assessment of Phase 7/8 propagation, though the signal remained weaker than anticipated—consistent with the Partial classification assigned to the MJO prediction in the February report.

3.5 Precursor Summary

All seven global precursor parameters assessed in February remained confirmed or intensified through March. The ENSO transition, stratospheric dynamics, and global SST fields evolved along the trajectories the framework identified. The most significant March development was

the acceleration of El Niño precursors, with the Niño 1+2 index reaching +1.5°C—substantially ahead of the pace implied by the framework's "mid-2026" emergence timeline. The downstream implications for April–May predictions (ME-1 dust sequence, SA-3 drought intensification, SEA heatwave) are assessed in the relevant regional sections.

4 Regional Verification

The following seven subsections assess each regional prediction set against observed March 2026 outcomes. Each region follows the identical structure established in the February report: predictions issued, observed events, verification assessment, and economic and human impact. Where predictions were previously assessed in February, the cumulative status is noted alongside the March-specific assessment.

4.1 Region 1: Australia

4.1.1 Predictions Under Assessment

Table 7: Region 1 — Predictions Under March Assessment

ID	Prediction	Prob.	$C(t)$	Window
AU-1	Tropical cyclone (Cat 3+), Queensland landfall	72%	0.71	Feb 15–Apr 5
AU-2	Late-season heatwave, SA/VIC (≥45°C)	78%	0.76	Mar 1–20
AU-3	Bushfire crisis continuation	80%	0.82	Feb–Mar
AU-4	Murray-Darling soil moisture deficit intensification	77%	0.74	Feb–Apr
AU-5	Delayed Autumn Break	82%	0.79	Apr–May

4.1.2 Observed Events

Tropical Cyclone Narelle (March 16–25+). Severe Tropical Cyclone Narelle formed in the Coral Sea on March 16 and underwent rapid intensification over waters at an all-time record 29.5°C, reaching Category 5 on the Australian scale by March 19 with sustained winds of 205–225 km/h and gusts to 285 km/h.

First landfall—Queensland, March 20: Narelle crossed the far north Queensland coast between Lockhart River and Cape Melville at approximately 07:00 AEST as a high-end Category 4 system, with sustained winds of approximately 195 km/h and gusts to 250–270 km/h. Rainfall totals reached 500 mm across Cape York Peninsula. Wenlock River gauge recorded 208 mm in 15 hours; Lockhart River Airport received 108 mm in 16 hours. BOM issued a dangerous storm tide warning for Princess Charlotte Bay. This was the most powerful cyclone to cross the Queensland coast since Cyclone Yasi in 2011 and the strongest to strike Cape York since Cyclone Mahina in 1899.

Second landfall—Northern Territory, March 21–22: After weakening to Category 2 over Cape York, Narelle re-intensified over the Gulf of Carpentaria and struck near Groote Eylandt as Category 3 with sustained winds of 148–165 km/h and gusts to 185 km/h. Rainfall of 200–230 mm in 24 hours fell across parts of the NT. Approximately 500 Numbulwar residents were evacuated by air to Darwin. Thirty Katherine Hospital patients, including nine pregnant women, were evacuated. Multiple boil water alerts were issued for East Arnhem communities.

Third landfall—Western Australia (in progress, March 25): The system emerged into the Indian Ocean north of Broome on approximately March 24 and re-intensified over 30°C waters. As of March 25, Narelle was Category 3 at 130 km/h sustained, located approximately 260 km NNW of Port Hedland, and forecast to strengthen to Category 4–5 (195–220 km/h) approaching North West Cape late March 26 with very destructive gusts to 260 km/h.

Narelle is only the 4th tropical cyclone to make triple mainland Australian landfall, compared most directly to Cyclone Ingrid (2005). Emergency declarations were activated across all three states and territories.

Katherine River Flooding (early March). A major flooding event separate from Cyclone Narelle struck the Northern Territory in early March. The Katherine River peaked at 19.2 m—the highest level in at least 20 years (since 1998). The levee was overtopped, inundating the main street. An Emergency Situation Declaration was issued for Katherine, Wugularr, and Daly River. 341 properties remained without power as of March 12.

Daly River Record Flood (March 14). The Daly River at Nauiyu peaked at 16.35 m, exceeding the previous all-time record of 16.25 m set in 1998. All residents were evacuated by air to Darwin—the second time this community was displaced by flooding in 2026.

South Australia Riverland Flooding (early March). The late-February/early-March low-pressure system brought 149 mm in 24 hours at Braemar Station (SA) and once-in-fifty-year rainfall to the Riverland region. A state of emergency was declared. Citrus and almond crop losses were estimated in the millions of AUD.

Murray-Darling Basin. Total MDB storage stood at 10,650 GL (48% capacity) as of March 5—19% (2,428 GL) lower than the same time the previous year. February rainfall brought significant improvement to soil moisture across southern agricultural regions (50–150 mm across much of Victoria and SA). BOM's autumn outlook projected below-average rainfall (60–80% chance) for most of Australia from April–June. Goulburn-to-Murray water trade remained closed.

Bushfire Season Transition. The catastrophic January Victorian bushfires (over 400,000 hectares, approximately 900 structures, 5 deaths) transitioned to recovery phase. Late-February/early-March widespread rainfall suppressed fire weather conditions through most of March. No new major bushfire events were reported. The AFAC autumn 2026 outlook noted increased fire risk for parts of NSW, VIC, SA, and SW WA.

4.1.3 Verification Assessment

Table 8: Region 1 — March Verification Assessment

ID	Prediction	Observed Match	Status	Cumul.
AU-1	Cat 3+ cyclone, QLD	Cat 5 peak; Cat 4 QLD landfall Mar 20; triple landfall QLD/NT/WA	Conf. (Exc.)	Partial → Conf.
AU-2	Heatwave SA/VIC	No heatwave; Riverland flooding; BOM: >80% above-avg temps Apr–Jun	Not Verif.	New
AU-3	Bushfire cont.	Jan season confirmed; Mar suppressed by rainfall; AFAC: elevated autumn risk	Confirmed	Conf. → Conf.
AU-4	MDB deficit	Storage 48% (19% below prior yr); Feb rain improved soil; Apr–Jun outlook dry	Confirmed	Conf. → Conf.
AU-5	Autumn Break	BOM: below-avg rainfall 60–80% Apr–Jun; El Niño risk reinforces	Developing	Devel. → Devel.

Assessment Notes. Prediction AU-1 represents the most significant reclassification in this report. The February verification classified it as Partial because TC Mitchell made landfall on the Western Australian coast rather than the predicted Queensland coast. Tropical Cyclone Narelle's Queensland landfall on March 20—within the original prediction window of February 15–April 5—confirms the prediction at a level substantially exceeding the specified threshold. The framework predicted Category 3+; Narelle peaked at Category 5 and delivered Category 4 at Queensland landfall. The additional NT and WA landfalls, producing only the 4th triple mainland landfall in Australian history, represent escalation beyond the prediction's scope.

The framework's coherence methodology correctly identified elevated cyclogenesis potential in the Coral Sea driven by record SSTs and active monsoon trough dynamics. The February Partial classification reflected incomplete data within a window that had 6+ weeks remaining—a conservative call that March data resolved definitively.

Prediction AU-2 is the report's only clean miss. The framework predicted a late-season heatwave in South Australia and Victoria during March 1–20. The observed conditions were instead characterized by significant inland rainfall and flooding. The subtropical ridge that sustained summer heat conditions through January–February broke down in late February, allowing moisture-laden systems to penetrate southern Australia. The predicted mechanism (persistent subtropical ridge, dry Murray-Darling soils amplifying sensible heat) was sound but the ridge did not persist through the specified window. The coherence value of 0.76 was the second-highest in the Australian prediction set, which—in contrast to the February report's finding that higher $C(t)$ correlated with higher confirmation rates—did not hold for this prediction. This suggests that ridge persistence may be less predictable via the coherence tensor than cyclogenesis or frontal dynamics.

4.1.4 Economic and Human Impact

Table 9: Region 1 — March Impact Summary

Metric	Observed
Fatalities	No confirmed fatalities from TC Narelle QLD/NT landfalls as of Mar 25; 0 from Katherine/Daly floods
Displacement	∼500 evacuated (Numbulwar); all Daly River residents evacuated (2nd time in 2026); Katherine levee overtopped
Infrastructure	341 properties without power (Katherine); banana farms destroyed (Cape York); Pilbara LNG/iron ore operations threatened (WA landfall pending)
Flooding records	Katherine River: 20-year high (19.2 m); Daly River: all-time record (16.35 m); Cairns: 258.6 mm/24 hr
Agricultural losses	Riverland citrus/almond losses in millions AUD; MDB storage 19% below prior year
Season context	2025–26 cyclone season: most active start since 1973–74 (6 named systems by Jan 6)

The absence of fatalities from Tropical Cyclone Narelle's Queensland and NT landfalls—despite Category 4–5 intensity—reflects the remote geography of Cape York and effective pre-positioning of evacuations in the NT. The pending WA landfall, with forecast Category 4–5 intensity targeting the industrially significant Pilbara coast, represents the highest remaining economic risk from this event.

4.2 Region 2: China

4.2.1 Predictions Under Assessment

Table 10: Region 2 — Predictions Under March Assessment

ID	Prediction	Prob.	$C(t)$	Window
CN-2	Severe cold wave sequence (≥3 events)	74%	0.72	Feb–Mar
CN-3	Major dust storm event (Beijing, NE China, Korean Peninsula)	75%	0.73	Mar 1–30
CN-5	Spring drought intensification, SE China	77%	0.75	Mar–May
CN-6	Tibetan Plateau heavy snowfall event	62%	0.60	Feb–Mar
—	Late cold surge causing crop damage	71%	0.81	Late Mar

4.2.2 Observed Events

Xinjiang Sandstorm (March 13). A large-scale sandstorm struck southern Xinjiang on March 13, mobilizing dust from the Taklamakan Desert across Hotan Prefecture. The NMC issued a blue-level sandstorm warning (lowest tier). Unconfirmed social media reports cited PM10 concentrations of 5,170 μg/m^3 in Hotan, though this value has not been officially verified. For context, the WMO notes severe dust storms commonly produce PM10 between 1,000 and 3,000 μg/m^3, while extreme events can exceed 5,000 μg/m^3. The event did not reach Beijing or the Korean Peninsula based on available reporting.

Cold-Air Outbreak (March 13). The same cold-air system that drove the Xinjiang sandstorm represented at least one additional cold surge beyond the four confirmed in February, bringing the cumulative total to a minimum of five cold wave events for the February–March window.

SE China Drought. NMC issued drought advisories in February warning that southern and southwestern China must "guard against developing drought" through spring. No March-specific quantitative precipitation deficit data for Fujian, Guangdong, or Zhejiang provinces were identified in English-language sources. IQAir reported on March 23 that Beijing, Wuhan, Hangzhou, and Guangzhou ranked among the most polluted major cities globally with AQI readings above 150, which may reflect dust and/or agricultural burning but is not directly indicative of drought severity.

4.2.3 Verification Assessment

Table 11: Region 2 — March Verification Assessment

ID	Prediction	Observed Match	Status	Cumul.
CN-2	Cold wave ≥3	4 confirmed Feb + ≥1 Mar = ≥5 total; threshold exceeded	Confirmed	Conf. → Conf.
CN-3	Dust storm, Beijing	Mar 13 Xinjiang confirmed; blue warning; did not reach Beijing or Korea	Partial	C. Early → Partial
CN-5	SE China drought	NMC advisory confirmed; quantitative Mar data unavailable in English	Developing	Devel. → Devel.
CN-6	Tibetan snowfall	Confirmed Feb (moderate-heavy Jan 31–Feb 6); no additional Mar data	Confirmed	Conf. → Conf.
—	Late cold surge	No reports of late-Mar cold surge affecting flowering crops	Not Verif.	New

Assessment Notes. Prediction CN-3 presents a nuanced classification challenge. The February dust storm of February 21–23—which struck 130+ cities with AQI 500, Level 14 gusts, and a Level-IV emergency response—was classified as Confirmed (Early) because it arrived before the

March 1–30 window. The March 13 Xinjiang sandstorm falls within the specified window but was geographically limited to southern Xinjiang and did not reach Beijing or the Korean Peninsula as predicted. Taken together, the prediction identified the correct mechanism (Mongolian cyclogenesis, cold-front-driven dust mobilization), the correct season, and the correct hazard class, but the within-window event was less severe and more geographically confined than predicted.

The cumulative classification reflects this: the February event confirmed the mechanism and impact class but arrived early; the March event confirmed the timing but not the geographic extent. The combined assessment is Partial for the March standalone (geographic deviation) while the cumulative record demonstrates the framework correctly identified the 2026 spring dust storm season.

Honest Framing

The China regional assessment is the weakest in this report due to systematic data limitations. CMA and NMC operational products—including drought monitoring, dust storm tracking, and cold wave advisories—are primarily published in Chinese. English-language availability is limited and delayed, typically by weeks to months. The CN-5 drought assessment in particular cannot be meaningfully evaluated without access to Chinese-language NMC drought monitoring pages. Researchers with Chinese-language capability are encouraged to verify these assessments against primary CMA/NMC sources. This limitation is structural and will affect all subsequent reports in this series.

4.2.4 Economic and Human Impact

Table 12: Region 2 — March Impact Summary

Metric	Observed
Fatalities	No consolidated figure available for March weather events
Air quality	Hotan PM10 reportedly 5,170 μg/m^3 (unverified); Beijing/Wuhan/Hangzhou/Guangzhou AQI >150 (Mar 23)
Agricultural exposure	SE China drought advisory active; spring planting season at risk pending Mar–Apr rainfall
Dust impact	Mar 13 event confined to southern Xinjiang; transportation disruption in Hotan Prefecture
Cumulative context	Feb 21–23 dust storm: 130+ cities, AQI 500, Level-IV emergency; 9.5B Spring Festival trips disrupted

4.3 Region 3: United States

4.3.1 Predictions Under Assessment

Table 13: Region 3 — Predictions Under March Assessment

ID	Prediction	Prob.	$C(t)$	Window
US-1	California AR sequence (Cat 3–4), multi-wave	81%	0.80	Feb 10–28
US-5	Western snowpack deficit persistence (<75% of normal)	67%	0.65	Feb–Mar (season)
US-7	Severe weather outbreak (tornadic), Gulf States / SE	64%	0.62	Feb–Mar
—	Spring tornado outbreak (Tornado Alley / Dixie Alley)	73%	0.69	Apr 1–30

4.3.2 Observed Events

Tornado Outbreak of March 5–7 (30 confirmed tornadoes, 8 fatalities, 29+ injuries). A powerful early-March storm system unleashed a significant and deadly tornado outbreak across the Central and Midwestern United States. The Storm Prediction Center had issued a Moderate Risk for severe weather across parts of Oklahoma, Kansas, and northern Texas.

The deadliest tornado of the first wave struck near Union City in Branch County, Michigan as an EF3 with peak winds of 160 mph. NWS Northern Indiana (IWX) confirmed this as the earliest EF3-or-stronger tornado in Michigan's recorded calendar year and the strongest to strike the state in 49 years (since an F4 struck Kalamazoo on April 2, 1977). Three persons were killed and 12 injured. The same supercell's earlier tornado at Three Rivers in St. Joseph County reached EF2 intensity (130 mph), injuring 110 persons when it struck a Menards store and Dollar Tree—the most injurious single tornado of the outbreak. One fatality was reported in Cass County.

In Oklahoma, the first wave produced an EF2 near Helena (Alfalfa County) that killed a 47-year-old woman and her 13-year-old daughter when their vehicle was struck. An EF3 tornado (135–140 mph, 6.8-mile path) struck the city of Beggs in Okmulgee County, killing two and causing severe damage to Beggs Middle School and High School. Tornadic activity continued overnight into Texas and Arkansas, with an EF2 near Willisville, Arkansas injuring one person.

No tornado watch was in effect for Michigan during the outbreak. Governor Whitmer issued a state of emergency in Branch, St. Joseph, and Cass counties and called for a probe into the absence of a tornado watch, questioning whether NWS funding cutbacks contributed to the gap.

Tornado Outbreak of March 10–11 (∼85 tornado reports, 4+ fatalities). A second major outbreak produced the month's most extreme event and one of the most significant tornadoes in recent U.S. history.

An **EF5 tornado struck Wells County, Indiana** with a 14.6-mile damage path and winds exceeding 200 mph—only the second EF5 in the United States since the El Reno, Oklahoma tornado of May 31, 2013. The event is described as "virtually unheard of" for early March. NWS IWX conducted extensive damage surveys confirming the rating.

Separately, a long-track supercell produced 12 confirmed tornadoes across Illinois and Indiana in a single storm. The most significant was an EF3 tracking from near Aroma Park in Kankakee County, IL northeast across the state line into Newton County, IN, striking Lake Village before continuing toward DeMotte. Peak winds reached approximately 150 mph across a path exceeding 40 miles over approximately 90 minutes. Ed and Arlene Kozlowski of Lake Village were killed when the tornado struck their home. NWS Chicago (LOT) documented the event as one of the most prolific single-supercell tornado producers in the region's history.

Record hail of 5–6 inches in diameter was reported in Kankakee—potentially a new Illinois state record, pending NWS verification.

March 13–17 Storm Complex. A large and powerful extratropical cyclone produced simultaneous severe weather and a historic blizzard—the most significant compound winter/severe event of 2026 to date.

Severe weather component: On March 15, SPC issued an Enhanced Risk across the Lower Mississippi Valley and a separate Enhanced Risk for Georgia and Alabama. On March 16, a **Moderate Risk (Level 4 of 5)** was issued for the Mid-Atlantic from northern South Carolina to southern Maryland, with 15% hatched tornado probability—a rare designation for the region. Seventy-eight wind damage reports of ≥58 mph gusts were documented across eastern North Carolina, Virginia, Maryland, and New Jersey. The squall line produced an EF2 in North Carolina. A total of 98 tornadoes were confirmed from the March 13–17 outbreak complex across all phases.

Blizzard component: The Upper Midwest received extraordinary snowfall totals reaching 3–4 feet (91–122 cm). Selected observations:

Table 14: March 13–17 Blizzard — Selected Snowfall Totals

Location	Total	Record Context	Source
Round Lake, MI (Upper Peninsula)	54 in (140 cm)	Storm maximum	NWS
Marquette, MI (NWS station)	36.3 in / 2 days	Surpassed 1997 record by ∼5 in	NWS MQT
Green Bay, WI	26.6 in	2nd highest on record (behind Mar 1888)	NWS GRB
Wausau, WI	23.4 in / day	Daily record; 30.2 in / 3 days	NWS
Seattle, WA (SeaTac)	3 in snow + 1.22 in rain	Both Mar 13 records	NWS SEW

The system's minimum central pressure reached 983 mb over Michigan before deepening to 944 mb over Quebec. States of emergency were declared in Wisconsin (Governor Evers, March 14) and Michigan (Governor Whitmer, 7 counties, March 17). Minneapolis and St. Paul declared snow emergencies. Over 500,000 customers lost power nationwide (140,000 in Michigan). More than 5,000 flights were cancelled by March 16. Minnesota State Patrol documented 464 crashes from March 13–15. At least 3 fatalities were attributed to the winter weather component.

Total March 2026 Tornado Count. Approximately 150 preliminary tornado reports were logged as of March 19—nearly double the ∼80 climatological average for March. Severe hail reports numbered approximately 330 and wind damage reports approximately 1,272. Combined with 48 February tornadoes, the January–March 2026 total substantially exceeded the seasonal average.

Western U.S. Heat Dome (March 12–25+). An extraordinary and prolonged heat dome established over the Western United States beginning approximately March 12–14, peaked March 18–22, and persisted through at least March 25. The event was described by the Associated Press as potentially "one of the most expansive heat waves in American history."

The heat dome was driven by a stationary ridge of high pressure with 500 mb heights reaching approximately 598 dm—3.5–4 standard deviations above normal. Temperatures ran 20–40°F above average across the Western and Central United States.

Table 15: Western U.S. Heat Dome — Selected Records

Location	Temperature	Record Context	Source
Winterhaven, CA	112°F (44.4°C)	All-time U.S. March record	NWS/NCEI
Yuma, AZ	112°F	Tied national March record	NWS
Phoenix, AZ	105°F	Earliest triple-digit day in history	NWS PHX
San Francisco, CA	89°F (Airport)	Surpassed 85°F record (1953); records since 1874	NWS
Boulder, CO	89°F	All-time March record; broken 3 consecutive days	NWS BOU
Denver, CO	86°F	All-time March record	NWS
Fort Collins, CO	91°F	Surpassed both March and April records (since 1893)	CSU
Lincoln, NE	97°F	Tied all-time April record—in March	NWS

Fourteen states set all-time March temperature records: California, Arizona, Nevada, Kansas, New Mexico, Nebraska, Utah, South Dakota, Missouri, Iowa, Colorado, Wyoming, Minnesota, and Idaho. NCEI registered 479 weather stations breaking March records during the peak week, with approximately 180 cities tying or setting new March records. The 112°F national reading was only 1°F below the all-time U.S. April record.

World Weather Attribution published a rapid analysis on March 21 concluding that the event had a return period of approximately 500 years even in the current warmed climate. Events of this magnitude have become approximately 4°C warmer and approximately 800 times more likely due to human-induced climate change. The event would have been "virtually impossible" without anthropogenic warming. March shows the most substantial warming signal for heat

extremes in this region compared to all other months.

Nebraska Wildfire Emergency. The heat dome and extreme fire weather produced the Morrill Fire (ignited March 12 northeast of Bridgeport, NE), which burned 643,074 acres—the largest single wildfire in Nebraska state history. The fire traveled approximately 70 miles in under 12 hours. One person died (Rose White, 86, attempting to evacuate). Additional Nebraska fires (Cottonwood: 128,036 acres; Road 203: 35,912 acres; Anderson Bridge: 17,400 acres) brought the total to over 824,865 acres, exceeding Nebraska's previous record year (2012). SPC issued an Extremely Critical fire weather risk designation. Governor Pillen described it as "one of the worst natural disasters in Nebraska history."

Western Snowpack. Snowpack conditions reached historic lows during March, compounded by the heat dome's accelerated melt:

- **Colorado:** Statewide SWE was the lowest on record in the SNOTEL era (∼1987) and the lowest since 1981 at manual snow course sites with 50+ years of data. At January 1, 38% of SNOTEL sites with 20+ years of data were at record-low snowpack. Upper Colorado River Basin: 57% of average. South Platte: 66%. Arkansas Basin: 47%.
- **California / Sierra Nevada:** DWR's February 27 survey measured 66% of average statewide (47% at Phillips Station). By March 24, California snowpack had plummeted to 34% of normal, with DWR warning snowpack was melting at approximately 1% per day. California could face the second-lowest April 1 snowpack in recorded history.
- **Broader West:** January 4, 2026 snow cover was the lowest January 4 in the MODIS satellite record (since 2001). Oregon, Colorado, and Utah all reported SNOTEL-era record low statewide snowpack as of February 1. Over 80% of SNOTEL stations in WA, OR, CO, UT, AZ, and NM experienced snow drought (SWE below 20th percentile). Bureau of Reclamation's most probable forecast showed Lake Powell reaching minimum power pool elevation by December 2026.

Climate scientist Daniel Swain (Weather West) stated: "April 1 snowpack may well be the worst on record across many, if not most, Western U.S. watersheds."

4.3.3 Verification Assessment

Table 16: Region 3 — March Verification Assessment

ID	Prediction	Observed Match	Status	Cumul.
US-5	Snowpack <75%	SNOTEL-era record lows; CA 34%; CO lowest since 1981; Lake Powell hydropower at risk	Conf. (Exc.)	Conf. → Conf. (Exc.)
US-7	Severe weather	∼150 Mar tornadoes (∼2× avg); EF5 IN (first since 2013); EF3 earliest MI; 12+ dead; Level 4 Mid-Atlantic	Conf. (Exc.)	Conf. → Conf. (Exc.)
US-1	CA AR sequence	Feb confirmed; Mar shifted to heat dome; no significant Mar ARs	Confirmed	Conf. → Conf.

Assessment Notes. The United States region produced the report's strongest escalation pattern. Both US-5 and US-7, already confirmed in February, intensified substantially during March.

Prediction US-5 specified Western snowpack below 75% of normal. The observed outcome—SNOTEL-era record lows across multiple basins, California at 34% by March 24, Colorado lowest since 1981, and Bureau of Reclamation projecting Lake Powell reaching minimum power pool—represents exceedance by a wide margin. The framework's coherence value of 0.65 (the

lowest assigned to any US prediction) and probability of 67% were the most conservative in the regional set, yet the prediction produced the most extreme observed outcome. This parallels the February finding for EU-3 (Alpine avalanches), where the lowest-probability European prediction produced the highest-severity outcome. The pattern suggests the framework may systematically underweight persistent ridge dynamics as a geometric constraint, calibrating conservatively on phenomena whose coherence signatures differ from synoptic-scale storm systems.

The March heat dome, while not explicitly predicted, is mechanistically coupled to the snowpack prediction. The framework identified the persistent North Pacific Ridge as a "high-coherence blocking configuration" that deflected the storm track north of California. That same ridge architecture, when amplified by the March SSW-driven pattern reconfiguration, produced the heat dome. The framework captured the underlying mode; the specific hazard manifestation (extreme heat rather than continued precipitation deficit) was not resolved. This represents a compound event class—where a predicted persistent mode produces an unpredicted secondary hazard—that warrants explicit treatment in v1.3 development.

Prediction US-7 specified severe weather outbreaks including tornadic activity in the Gulf States and Southeast. The observed March outcome—an EF5 tornado (the first since 2013), approximately 150 March tornadoes (double the climatological average), the earliest EF3 in Michigan's history, and a Level 4 Moderate Risk issued for the Mid-Atlantic—substantially exceeds the prediction's scope. The framework's probability of 64% and coherence value of 0.62 were again the lowest in the US prediction set, consistent with the pattern of conservative calibration on severe convective events.

The March 5–7 outbreak is particularly notable for its early timing. The framework's prediction specified a February–March window, which conventional forecasting would consider aggressive for EF3+ activity in Michigan. The observed EF3 in Union City (March 6) and EF5 in Wells County (March 10) confirm that the framework's assessment of anomalous baroclinic gradients—driven by the SSW-displaced polar vortex interacting with Gulf moisture return—correctly identified the conditions for early-season violent tornado potential.

Honest Framing

The Western U.S. heat dome of March 2026 was not predicted by the Auburn Coherence Framework. This event—which shattered March temperature records in 14 states, was assessed by World Weather Attribution as "virtually impossible" without climate change, and triggered the largest wildfire in Nebraska history—represents a significant unpredicted outcome within the framework's assessment domain. The framework's value proposition is geometric constraint identification, not comprehensive hazard enumeration. The heat dome's mechanistic linkage to predicted snowpack and drought dynamics suggests the framework captured the forcing mode while missing the specific hazard expression. This limitation is documented with equal prominence to the confirmed predictions, consistent with the honest framing commitment established in the February report.

4.3.4 Economic and Human Impact

Table 17: Region 3 — March Impact Summary

Metric	Observed
Fatalities	15+ (tornado outbreaks 12+; blizzard 3; wildfire 1)
Power outages	>500,000 customers (blizzard); additional outages from tornado damage
Flight cancellations	>5,000 (blizzard complex)
Wildfire	824,865 acres (Nebraska; largest in state history)
Snowpack	SNOTEL-era record lows; CA 34% of normal; Lake Powell hydropower at risk
Tornado records	EF5 (first since 2013); earliest MI EF3; ∼150 Mar tornadoes (∼2× average)
Heat records	14 states set all-time March records; 112°F national March record; 479 stations broke records
Emergency declarations	WI, MI (blizzard); NE (wildfire); MI (tornadoes)
Estimated economic losses	Multi-billion USD combined (wildfire, tornado damage, snowpack-driven water supply impacts); institutional estimates pending

The United States experienced the broadest range of simultaneous hazard types of any region: violent tornadoes (EF5), a historic blizzard (54 inches), a record-shattering heat dome (112°F), the largest wildfire in a state's history, and a snowpack crisis with implications for hydroelectric generation at Glen Canyon Dam. The temporal compression—all five hazard types occurring within a single 20-day window (March 5–25)—illustrates the compound event architecture the framework's coherence analysis was designed to identify: a single atmospheric configuration (SSW-driven meridional flow interacting with record ocean heat) producing multiple hazard modes simultaneously across different geographic sectors of the same continental domain.

4.4 Region 4: Western Europe

4.4.1 Predictions Under Assessment

Table 18: Region 4 — Predictions Under March Assessment

ID	Prediction	Prob.	$C(t)$	Window
EU-1	Catastrophic Iberian flooding (AR-driven)	84%	0.83	Feb 5–Mar 15
EU-2	SSW-driven cold outbreak (Beast from East analog)	68%	0.66	Feb–Mar
EU-3	Alpine avalanche cycle (Danger Level 4–5)	65%	0.63	Feb–Mar
EU-4	UK / Rhine compound riverine flooding	72%	0.70	Mar–Apr
EU-5	Scandinavian blocking persistence + negative NAO	80%	0.79	Jan–Feb

4.4.2 Observed Events

Iberian Storm Sequence Continuation (March 1–25). The extraordinary 2025–26 European windstorm season continued through March with reduced but persistent intensity. Storm Regina (∼March 1–4), the 17th named storm of the season, equaled the all-time record for most named storms in a single season—a threshold typically reached in mid-April, not early March. AEMET issued yellow and orange warnings across 57 forecast areas, with wind gusts to 70 km/h and orange rainfall warnings in Almería for 15 mm/hr intensity. Storm Therese (March 19–22) delivered heavy rainfall exceeding 300 mm forecast on La Palma and Tenerife, with gusts above 90 km/h and waves exceeding 5 m. Tenerife activated its Island Emergency Plan. By March

25, 19 named storms had occurred in the 2025–26 season—an unprecedented pace. No updated consolidated damage estimate beyond the €4 billion February figure was published.

Saharan Dust "Blood Rain" Event (March 1–9). A major Saharan dust outbreak was lofted from northwest Africa around February 27 and transported northward across Europe from approximately March 1–9. The event coincided with Storm Regina's cut-off low positioning, which drew dust across Iberia and France into the UK.

NASA GEOS modeling and Suomi NPP satellite observations documented the plume, with AERONET stations recording elevated aerosol optical depth across the Mediterranean corridor. The Copernicus Atmosphere Monitoring Service (CAMS) confirmed PM10 concentrations exceeding EU air quality standards across multiple countries. At the University of Reading (UK), Dr. Claire Ryder's measurements showed coarse particle concentrations quadrupled during the event. Countries affected included Spain, Portugal, France, the UK, Switzerland, Italy, Belgium, the Netherlands, western Germany, and Ireland.

Research published in *Solar Energy* (Varga et al., 2026) using NASA MERRA-2 and MODIS data documented that photovoltaic energy generation in Hungary dropped to 46% efficiency on high-dust days versus 75%+ on low-dust days—an operationally significant finding for European renewable energy infrastructure.

UK Winter Rainfall and Groundwater Flooding. Winter 2025–26 was England's 8th wettest on record (since 1836). Southern England experienced its 4th wettest winter. West Midlands, Cornwall, and Leicestershire set all-time wet records. Southwest England received the 2nd highest December–February rainfall since 1871. February England rainfall reached 170% of the long-term average. Multiple stations logged 40 consecutive wet days (December 31–February 8).

March brought a marked shift. England received only 53% of the long-term average rainfall through March 23 (Environment Agency / National Drought Group, March 23). Despite the dry March, groundwater flooding persisted as of March 24, with the Environment Agency warning of "ongoing local flooding in parts of the South of England from groundwater for the next five days." West Sussex communities experienced springs, road flooding, and sewage network issues, with minor groundwater impacts expected to continue until April. Groundwater levels at Hanthorpe in Lincolnshire Limestone reached the highest February level since records began in 1972. Reservoir storage was at 94–95% for England. Ninety-one flood warnings were active across England and Wales during peak periods.

No significant March Rhine flooding was identified. The spring flood risk window remains open through April.

Alpine Avalanche Crisis Continuation (March). Avalanche conditions persisted and intensified through March. The European season total reached approximately 127 avalanche fatalities as of mid-March—the highest in 7–8 years and well above the ∼100/season average.

Key March incidents:

- **La Grave, France (March 15):** Fatal accident amid widespread avalanche activity across multiple French massifs. Crown fractures ranged from 20 cm to 1 m. Multiple rescue operations were conducted simultaneously across the Hautes-Alpes and Savoie departments.
- **Switzerland (March 16):** Danger reached Level 4 (High) in parts of the Swiss Alps after more than 1 m of new snow in 48 hours. SLF issued regional advisories for Valais and Bernese Oberland.
- **Val Ridanna, South Tyrol (March 21):** Two ski tourers were killed and five injured when 25 people were caught in an avalanche at 2,445 m elevation—despite only a Level 2 "Moderate" danger rating from Meteo Trentino. Six helicopters and approximately 80 rescuers responded. The event underscores the persistent weak layer problem identified in the February report: avalanche fatalities occurring at lower-than-expected danger ratings

indicate a weak layer architecture that standard danger-level communications do not adequately convey to recreational users.

Late-March Arctic Plunge (developing, March 22–28). Driven by the confirmed March SSW event and developing Greenland blocking, a dramatic pattern reversal was forecast for late March across Western and Central Europe. Severe Weather Europe (March 22) described a "Great March Reversal" with temperature drops of 15–20°C expected within two days. Widespread late-season snow was forecast to low elevations across France, Germany, and the Alpine forelands. Dangerous frost threatened vineyards and orchards that had already begun budding during the warm early-to-mid March period—a compound agricultural risk (warm spell inducing early phenological development followed by hard freeze) that mirrors the SSW-driven "Beast from the East" mechanism the framework predicted, albeit manifesting later and through a different blocking geometry than the 2018 analog.

4.4.3 Verification Assessment

Table 19: Region 4 — March Verification Assessment

ID	Prediction	Observed Match	Status	Cumul.
EU-1	Iberian flooding	Storm sequence continued; 19 named storms (record pace); window closed Mar 15	Confirmed	Conf. → Conf.
EU-2	SSW cold (BftE)	Late-Mar "Great Reversal"; Greenland blocking developing; cold now routing to W. Europe	Partial	Partial → Partial
EU-3	Alpine aval. 4–5	127 fatalities (7–8 yr high); Level 4 Switzerland Mar 16; Val Ridanna 2 dead	Confirmed	Conf. → Conf.
EU-4	UK/Rhine flood	UK: 8th wettest winter; groundwater persistent thru Mar 24; Rhine: no event	Partial	New
EU-5	Scandi. block + NAO	Winter blocking confirmed; NAO positive early Mar; reverting neg. late Mar	Confirmed	Conf. → Conf.

Assessment Notes. Prediction EU-1's window closed on March 15. The storm sequence continued through that date (Storm Regina, early March) and the season subsequently produced additional storms (Therese, March 19–22), though these fell outside the specified window. The 19 named storms by March 25 represent an unprecedented pace for the European windstorm season. The prediction is confirmed on the basis of the February–early March events within the specified window; the continued activity beyond the window reinforces the framework's identification of the persistent negative NAO/southward jet displacement mechanism.

Prediction EU-2 remains classified as Partial, though the late-March pattern evolution is noteworthy. The February assessment noted that the SSW mechanism was correctly identified but the cold was routed toward North America rather than Western Europe. The March SSW event, combined with developing Greenland blocking, is now directing cold air into Western Europe—precisely the mechanism the framework predicted. The timing (late March rather than February–early March) and blocking geometry (Greenland high rather than Scandinavian retrogression) differ from the 2018 "Beast from the East" analog, but the fundamental mechanism chain (SSW → blocking → cold air displacement into Western Europe) is verifying with a temporal lag. The Partial classification is maintained because the primary impact window specified in the prediction has passed; the late-March cold event represents a delayed manifestation of the predicted mechanism.

Prediction EU-3 continues to produce one of the report's most significant findings. The February report noted that the framework's lowest-probability European prediction (65%) produced the highest-severity outcome (Level 5, first in 17 years). March data reinforces this pattern: the season total of 127 fatalities exceeds the ∼100 average, avalanche danger

reached Level 4 in Switzerland, and the Val Ridanna event—25 caught, 2 killed at only Level 2 danger rating—highlights the persistent weak layer architecture the framework identified. The discrepancy between low assigned probability and extreme observed outcome continues to warrant investigation for v1.3 calibration.

Prediction EU-4 is classified as Partial in its first assessment month. The UK component has partially verified: England's 8th wettest winter on record, persistent groundwater flooding through March despite a dry month, and record-high groundwater levels all confirm the saturated-soil/continued-flooding mechanism. The Rhine component has not verified during March. The Mar–Apr window remains open; spring thaw flooding driven by Alpine snowmelt (which the late-March cold spell may delay) could still produce Rhine impacts in April.

Saharan Dust Event — Cross-Regional Significance. The March 1–9 Saharan dust outbreak, while not explicitly predicted, intersects with multiple framework themes: the persistent negative NAO/southward jet that steered both Atlantic storms and Saharan air masses into Iberia, the anomalous North African thermal gradient, and the renewable energy infrastructure vulnerability to aerosol loading. The documented 46% efficiency reduction in Hungarian photovoltaic generation during peak dust days represents a quantifiable economic impact of trans-continental aerosol transport—a hazard class not captured by conventional weather prediction but potentially identifiable through the framework's geometric constraint analysis of jet stream positioning.

4.4.4 Economic and Human Impact

Table 20: Region 4 — March Impact Summary

Metric	Observed
Fatalities	5+ (Alpine avalanches: 2 Val Ridanna, 1+ La Grave, additional incidents)
Avalanche season total	∼127 fatalities (highest in 7–8 years; above ∼100 average)
Storm season	19 named storms by Mar 25 (unprecedented pace; record equaled in early Mar)
UK flooding	8th wettest winter on record; groundwater flooding persistent; 91 warnings active
Dust impact	PM10 exceeding EU standards across 10+ countries; PV efficiency drop to 46%
Cumulative Iberian losses	>€4 billion (Feb estimate; Mar update pending)
Late-Mar frost risk	Vineyards and orchards at risk from hard freeze after early budding

4.5 Region 5: Middle East

4.5.1 Important Context — Conflict-Degraded Observation Capacity

Data Limitation Notice

The U.S.-Israeli-Iran conflict, initiated on February 28, 2026, has severely disrupted regional meteorological observation infrastructure. Iraqi, Kuwaiti, Iranian, and Syrian airspace has been closed. Ground-based weather stations, radiosonde launches, and satellite data downlink facilities in conflict zones are operating at degraded capacity or are offline. Emergency response coordination across the region is compromised. These conditions limit the availability and reliability of March 2026 meteorological data for the Middle East. Assessments in this section reflect the best available information under these constraints; data gaps are noted explicitly.

4.5.2 Predictions Under Assessment

Table 21: Region 5 — Predictions Under March Assessment

ID	Prediction	Prob.	$C(t)$	Window
ME-1	Severe dust storm sequence (Iraq, Gulf States)	88%	0.86	Mar–May
ME-2	Flash flooding from Red Sea Trough / jet superposition	62%	0.60	Mar–Apr
ME-3	Fertile Crescent drought persistence	85%	0.84	Ongoing
—	Early intense heat buildup (>50°C)	71%	0.68	Apr–May

4.5.3 Observed Events

Oman Flash Floods — "Al-Musarrat" System (March 20–24+). A low-pressure system designated "Al-Musarrat" by Omani authorities began affecting Oman on March 20, with impacts forecast to persist through March 30. As of March 23, 10 people had died including 5 children. Fatalities included:

- Two persons swept by flash floods in Wadi Al Maawil
- Three persons from a vehicle carrying 10 people swept into floodwaters in Barka (including 2 Kerala nationals)
- Four children killed in a collapsed mud-brick building in Rustaq
- A 7-year-old drowned in Barka

At least 29 persons were rescued, 106 were displaced to a school in Adam, and more than 200 homes were damaged or destroyed. Downdraft winds reached 55–83 km/h with some reports of 155 km/h gusts. All governorates were affected. On March 25, most of the country remained under red and orange alerts from the Civil Aviation Authority (CAA) and Civil Defence and Ambulance Authority (CDAA). GDACS/ECHO maintained monitoring throughout.

March 25–31 Storm System (Active/Developing). A large-scale, slow-moving extratropical system described by the Washington Post (March 24) as a "Tornado Alley-style storm system" began affecting the broader Middle East on March 25. The atmospheric configuration was characterized as "virtually unheard of" for the region.

Key synoptic features identified by multiple analysis sources:

- CAPE values of 4,000–5,000 J/kg across the Gulf region (analysis by Tomer Burg)
- Low-level jet exceeding 70 km/h providing moisture transport
- Surface cyclone deepening to ∼995 hPa near Baghdad
- Exceptional moisture convergence from the Mediterranean and Red Sea
- Upper-level trough interaction with subtropical jet—the "jet superposition" mechanism specified in prediction ME-2

Model guidance projected widespread rainfall exceeding 100 mm across Iraq, with localized totals of 150–200 mm (ECMWF/GFS consensus) and extreme GFS scenarios reaching 500 mm in western Iran. Dubai was forecast to receive 75–150 mm (3–6 inches)—potentially a full year's annual rainfall (annual average ∼100 mm). For comparison, the April 2024 UAE event deposited 164 mm (6.45 inches) at Dubai International Airport, causing more than 1,200 flight cancellations and widespread infrastructure damage.

Supercells and tornadoes "cannot be ruled out" across the UAE, Qatar, eastern Saudi Arabia, and northern Oman on March 26–27 (The Watchers, MyRadar Weather). MyRadar Weather described a "rare desert tornado risk" including in Dubai.

Emergency preparations across the region included UAE National Centre of Meteorology (NCM) and National Emergency Crisis and Disasters Management Authority (NCEMA) activation; Qatar Labour Ministry weather alert (March 25); and Oman's pre-existing emergency operations under the Al-Musarrat system. Baghdad faces acute flood risk given infrastructure fragility from decades of underinvestment and severely degraded emergency response capacity from the ongoing military conflict. Iraqi airspace has been closed since February 28.

Additional: "Black Rain" Over Tehran (March 8–9). A conflict-caused environmental weather event occurred when Israeli airstrikes on March 7–8 struck multiple oil facilities in and around Tehran. A subsequent storm system produced toxic, oil-contaminated precipitation ("black rain") across Tehran (∼10 million population) and up to 70 miles northward. WHO warned of a "massive release" of toxic substances including hydrocarbons, PM2.5, polycyclic aromatic hydrocarbons (PAHs), sulfur dioxide, and heavy metals. While not a natural weather event, the interaction between military-caused atmospheric contamination and natural precipitation systems represents a compound hazard class with implications for the framework's "dust-to-flood cascade" scenario described in the original prediction document.

Dust Storm Activity. No standalone major dust storm events were specifically confirmed for March 1–25 in available English-language reporting. This absence may reflect genuinely reduced early-season activity or degraded observation and reporting capacity due to the military conflict. Haboobs (convective dust storms) are forecast as a secondary hazard from the March 25–31 storm system. The primary March–May dust storm window remains open.

Fertile Crescent Drought. The multi-year drought continues as the worst since 1933. Lake Tharthar stood at 38.5 m above sea level in October 2025—the lowest since the reservoir's 1958 construction and more than 20 m below its 1993 peak. Lake Habaniya held only 511 million m^3 (15.5% of capacity). National water reserves reached an 80-year low of approximately 4 billion m^3. Tigris and Euphrates flow has been reduced up to 27%, affecting approximately 8 million people. The UNESCO-designated Mesopotamian Marshes have lost approximately 84% of permanent area. Iraq's Ministry of Agriculture suspended September 2025 farming plans including wheat cultivation. No specific data on the impact of February's record rainfall on reservoir levels was available at the time of writing; the March 25–31 storm system could deliver significant inflow if precipitation projections verify.

4.5.4 Verification Assessment

Table 22: Region 5 — March Verification Assessment

ID	Prediction	Observed Match	Status	Cumul.
ME-1	Dust sequence	No confirmed major standalone events; observation capacity degraded; window open	Developing	New
ME-2	Flash flood, jet sup.	Oman: 10 dead, 200+ homes; Mar 25–31: 100–500 mm Iraq/Iran; mechanism confirmed	Confirmed	C. Early → Conf.
ME-3	Fertile Crescent	Worst since 1933; Lake Tharthar at construction-era low; 84% marsh loss	Confirmed	Conf. → Conf.
—	Heat >50°C	Window Apr–May; no precursor data identified	Developing	New

Assessment Notes. Prediction ME-2 upgrades from Confirmed (Early) to Confirmed. The February report classified the Syria floods (February 7–12) and Iraq record rainfall as Confirmed (Early) because they arrived 3–4 weeks ahead of the specified March–April window. The March events—Oman's Al-Musarrat flooding (10 fatalities, March 20–24) and the March 25–31 storm

system now affecting the broader region—fall squarely within the predicted window. The atmospheric mechanism matches the prediction with high fidelity: the March 25–31 system is driven by upper-level trough interaction with the subtropical jet over the region, producing the "jet superposition" configuration the framework identified. The CAPE values of 4,000–5,000 J/kg and surface cyclone near Baghdad represent the thermodynamic and dynamic ingredients the framework's Red Sea Trough interaction scenario described.

The March 25–31 event, if precipitation projections verify, may exceed the April 2024 UAE event that serves as the framework's primary analog. The Washington Post's characterization of the atmospheric setup as "virtually unheard of" is consistent with the framework's assessment that jet superposition events in this region exhibit high coherence ($C(t) = 0.77$ in the original prediction, classified HIGH) and geometric persistence once established.

Prediction ME-1 cannot be meaningfully assessed due to degraded observation capacity. The framework assigned this prediction the highest probability (88%) and highest coherence value (0.86) of any prediction in the Middle East set. The March–May dust storm window is the primary seasonal period for Shamal-driven events; the absence of confirmed events in March reporting likely reflects data limitations rather than non-occurrence. This assessment will be revisited if observation capacity improves.

Honest Framing

The Middle East regional assessment operates under the most severe data constraints of any region in this report. The ongoing military conflict has disrupted the meteorological observation infrastructure, emergency response coordination, and institutional reporting capacity upon which verification depends. Predictions ME-1 and the heat buildup scenario cannot be fairly assessed under these conditions. The Developing classification for ME-1 should not be interpreted as evidence against the prediction; it reflects the inability to verify rather than the absence of the predicted phenomenon.

4.5.5 Economic and Human Impact

Table 23: Region 5 — March Impact Summary

Metric	Observed
Fatalities	10+ (Oman Al-Musarrat flooding, including 5 children); Mar 25–31 event impacts pending
Displacement	106 displaced (Oman/Adam); additional displacement expected from Mar 25–31 system
Infrastructure	200+ homes damaged/destroyed (Oman); Baghdad at acute flood risk
Drought severity	Worst since 1933; Lake Tharthar at 1958-construction low; national reserves at 80-yr low
Environmental	"Black rain" over Tehran (10M population) from conflict-caused oil facility strikes
Mar 25–31 forecast	Models: 100–500 mm Iraq/Iran; year's rainfall in 3 days for Dubai; tornado risk in Gulf States

The convergence of natural weather hazards with active military conflict across the Middle East represents a compound risk class that no climate prediction framework can fully capture. The March 25–31 storm system is arriving during a period of closed airspace, degraded infrastructure, displaced populations, and compromised emergency response—conditions that amplify the human impact of any precipitation event regardless of its meteorological magnitude. The framework's identification of this period as one of elevated flash flood risk, issued 49 days before the current storm system, underscores the potential value of extended-range geometric constraint

identification in conflict-affected regions where operational forecast lead times are measured in hours rather than weeks.

4.6 Region 6: South America

4.6.1 Predictions Under Assessment

Table 24: Region 6 — Predictions Under March Assessment

ID	Prediction	Prob.	$C(t)$	Window
SA-1	SE Brazil flooding / landslides (SACZ-driven, Minas Gerais / São Paulo)	79%	0.78	Feb–Mar
SA-2	Coastal El Niño flooding (Peru / Ecuador)	66%	0.64	Mar–Apr
SA-3	Agricultural drought, Argentine Pampas (corn/soy impact)	74%	0.72	Mar–May
SA-C	Compound: South American Dipole — simultaneous SA-1 + SA-2 + SA-3	—	—	Feb–Apr

4.6.2 Observed Events

SE Brazil — Zona da Mata Final Toll and March Continuation. The February Zona da Mata disaster reached a final confirmed toll of 73 fatalities (65 in Juiz de Fora, 7 in Ubá, 1 additional recovered in March; last body recovered March 17). Approximately 10,000 residents were displaced. INMET data showed Juiz de Fora received 752.4 mm in February—more than 4× the 170 mm monthly average and the wettest February ever recorded at that station (INMET series since 1961). World Weather Attribution conducted a rapid attribution study linking the event to climate change and unplanned urbanization.

SACZ-driven flooding continued into March. On March 8, heavy thunderstorms across São Paulo city generated 180 flooding-related emergency calls in 5 hours. Multiple streams overflowed (Moinho Velho, Ipiranga, Guaratiba, Mooca), with flooding depths reaching 170 cm in some areas. INMET and civil defense issued warnings for unstable conditions persisting through March 9–11. OCHA confirmed broader impacts extending to Lajes do Muriaé and Nova Iguaçu (Rio de Janeiro state, 681 persons sheltering) and Peruíbe (São Paulo state, 357 affected). GDACS flagged ongoing Brazil flood events as of March 23.

Peru — Coastal El Niño Emergency Escalation. The Coastal El Niño emergency, which the February report classified as Confirmed (Early) based on the ENFEN alert activation and 68 fatalities, escalated dramatically through March as the prediction's primary window (March–April) engaged.

ENFEN maintained its Alerta de El Niño Costero through March (Comunicado No. 05-2026), extending the projected duration through December 2026 (from November in the previous communiqué). The weekly Niño 1+2 index stood at +1.5°C as of March 11—reflecting substantial coastal warming even as the central Pacific Niño 3.4 approached neutral (−0.0°C). ENFEN explicitly noted that 2026 impacts do not yet compare to the catastrophic 2017 Coastal El Niño (1.7 million affected, 162 deaths, $3.12 billion damage).

The human toll escalated across reporting periods:

Table 25: Peru El Niño Costero — Escalation Timeline

Date	Source	Deaths	Districts in Emergency
Late February	OCHA	68	246
March 13	OCHA	58 (season count reset)	283
March 20	INDECI	74	1,177
March 23	OCHA	92	1,177

The discrepancy between OCHA and INDECI fatality counts likely reflects different counting methodologies and reporting windows. The critical figure is the emergency district count: 1,177 districts—**62% of all districts nationwide**—were under state of emergency declarations as of March 20, up from 283 on March 13 and 246 in late February. This represents a 379% increase in 7 days.

INDECI's March 20 situation report provided the most comprehensive damage assessment: 27,555 *damnificados* (severely affected), 196,483 total affected persons, 1,257 homes destroyed, 10,437 homes uninhabitable, and 92,980 homes damaged. Approximately 931 km of roads were damaged and 6,000 hectares of farmland destroyed, with estimated costs of 291 million soles (∼$84 million USD) as of early March. Most affected departments included Apurímac (12 deaths), Cusco (8), Piura (8), and Arequipa (7).

On March 22, a landslide killed 2 people in Machu Picchu district, damaging houses, roads, and a bridge, with heavy rainfall forecast to continue for 96+ hours.

Ecuador — National State of Emergency. Ecuador's rainy season escalated to a national crisis during March. The government declared a national state of emergency on March 12 (60 days, effective across all 24 provinces). Ecuador also requested EU Civil Protection Mechanism assistance on March 6, with Copernicus Emergency Management Service activated March 11, producing 16+ analytical maps of flood extent.

By the OCHA March 23 report, Ecuador had experienced 2,046 rain-related incidents since January, affecting 71,309 people, displacing 6,066, and killing 14. Damage included 19,420 homes, 8,350 hectares of crops destroyed, and more than 126,000 livestock killed. GDACS/ECHO documented 772 flood events and 710 landslides year-to-date, with 22 rivers simultaneously overflowing. Sentinel-1 satellite imagery from March 3 showed extensive flooding along the Babahoyo River.

Argentine Pampas — Partial Relief. Mid-February rainfall provided critical relief during reproductive crop stages, with March continuing the improved moisture regime. The Buenos Aires Grain Exchange (BAGE) reported on March 5 that 74% of soybeans and 87.4% of corn were in Normal/Excellent condition—a substantial recovery from February's 44% and 40% good-to-excellent readings. The corn production forecast was raised 1% to 56.7 MMT on March 20 based on above-average precipitation and cool temperatures. Soybean production was estimated at 48.5 MMT. Corn harvest advanced to 13% complete. Satellite-derived NDVI values were above-median in Córdoba (except Marcos Juárez), NW Buenos Aires, La Pampa, and Santa Fe.

However, Argentina's Ministry of Agriculture published Ministerial Resolution No. 88, declaring a **State of Agricultural Emergency and/or Agricultural Disaster from March 1 through August 31, 2026** for affected producers—confirming the drought's severity warranted formal declaration despite recent improvement. The northeastern Pampas remained relatively warm and dry, and long-term drought signals persisted.

Paraná River. The Paraguay River at Asunción was reported at 0.50 m above the zero hydrometric mark—a critical stage affecting navigation—declining 4–5 cm/day. The Naval Prefecture reduced vessel loading capacity due to low levels.

4.6.3 Verification Assessment

Table 26: Region 6 — March Verification Assessment

ID	Prediction	Observed Match	Status	Cumul.
SA-1	SE Brazil SACZ	73 dead (final); SACZ flooding continued Mar 8 SP (180 emergency calls); OCHA: ongoing Rio/SP	Confirmed	Conf. → Conf.
SA-2	El Niño Costero	Peru: 92 dead, 1,177 districts (62%) emergency; Ecuador: nat'l emergency Mar 12; Niño 1+2 +1.5°C	Conf. (Exc.)	C. Early → Conf.
SA-3	Pampas drought	Relief to 74–87% normal; but Ministry declared Agric. Emergency Mar 1–Aug 31	Confirmed	Conf. → Conf.
SA-C	S. Am. Dipole	All three legs simultaneous through Mar: Brazil flooding + Peru/Ecuador emergency + Argentine drought declaration	Confirmed	Conf. → Conf.

Assessment Notes. South America again produced the framework's cleanest regional verification: four predictions (including the compound risk), four confirmed, zero partial or not verified. This result extends the February finding of "three predictions, three confirmed" to a two-month record of sustained verification.

Prediction SA-2's reclassification from Confirmed (Early) to Confirmed reflects the primary window (March–April) now being fully engaged. The February report classified it as Early because ENFEN's alert activation on February 13 and the escalating fatality count preceded the specified window. March data removes this temporal caveat entirely: the prediction's March–April window is now producing impacts that substantially exceed the February precursors. The expansion from 283 to 1,177 emergency districts in 7 days (March 13–20) represents the most dramatic single escalation in any prediction across this report.

The framework's identification of coastal SST decoupling—where Peru's littoral zone warmed +2 to +3°C while the basin-wide Niño 1+2 index read 0.0°C in February—was operationally validated. By March, the Niño 1+2 index had risen to +1.5°C, confirming the eastward propagation of the warming signal the framework identified as a critical precursor. This diagnostic capability—distinguishing coastal from basin-wide ENSO signals—is not captured by standard ENSO monitoring indices and represents a differentiating feature of the geometric constraint approach.

Prediction SA-3 presents an interesting interpretive case. The February drought stress (corn at 44% good-to-excellent, soybeans at 40%) confirmed the prediction within its specified window. March rainfall provided substantial relief, with conditions recovering to 74–87% Normal/Excellent. A conventional assessment might downgrade this to Partial on the basis that the drought did not persist through March. However, two factors support maintaining the Confirmed classification: (a) the prediction specified a March–May window, and the drought was confirmed within that window during its opening month; and (b) Argentina's Ministry of Agriculture declared a formal State of Agricultural Emergency effective March 1–August 31, confirming at the institutional level that the drought's severity warranted extraordinary measures despite recent improvement. When the government declares an agricultural disaster during a month of apparent recovery, the prediction's identification of structural drought stress is vindicated.

Compound Risk 3 — South American Dipole. The compound scenario continues to verify across all three legs simultaneously:

- **Tropical Atlantic excess (SA-1):** SACZ-driven flooding continued into March across Minas Gerais, São Paulo, and Rio de Janeiro states.
- **Eastern Pacific coastal warming (SA-2):** Niño 1+2 at +1.5°C; Peru and Ecuador in escalating emergency.

- **Subtropical continental deficit (SA-3):** Despite March rainfall relief, Argentina declared formal agricultural emergency; Paraná River at critical navigation levels.

The framework's February prediction document stated that these three hazards are "geometrically linked through the South American monsoon system's moisture transport architecture." Two months of simultaneous materialization across all three legs continues to support this assessment. Standard independence-based probability models would assign a joint probability of $0.79 \times 0.66 \times 0.74 = 0.386$ (approximately 39%) to all three co-occurring. The framework's geometric coupling argument—that SACZ stalling diverts moisture from the Pampas while Walker circulation anomalies reinforce coastal warming—has been sustained across the full February–March period without interruption.

4.6.4 Economic and Human Impact

Table 27: Region 6 — March Impact Summary

Metric	Observed
Fatalities	95+ cumulative (Peru 92, Brazil 73 final from Feb + ongoing Mar, Ecuador 14)
Displacement	23,000+ (Peru); 10,000 (Brazil, Zona da Mata); 6,066 (Ecuador)
Emergency declarations	Peru: 1,177 districts (62% nationwide); Ecuador: national emergency (all 24 provinces); Argentina: Agricultural Emergency Mar 1–Aug 31
Agricultural losses	Peru: 6,000 ha farmland destroyed, 931 km roads damaged; Ecuador: 8,350 ha crops destroyed, 126,000+ livestock killed; Argentina: corn forecast revised (recovery)
Infrastructure	Peru: $84M estimated costs (early Mar); Ecuador: 19,420 homes damaged; Paraná: critical navigation levels
Cumulative Feb–Mar	Combined regional fatalities >180; three countries under simultaneous emergency declarations

South America recorded the highest cumulative fatality count of any region across the February–March period. The concentration of casualties in developing-nation contexts—informal settlements on unstable slopes in Minas Gerais, under-resourced emergency response in Peruvian Andean communities, and riverine communities across Ecuador's coastal lowlands—continues to underscore the framework's potential utility in pre-positioning disaster response resources in regions where operational lead time translates most directly to lives saved.

The simultaneous declaration of emergency status across three South American countries—Peru (1,177 districts), Ecuador (all 24 provinces), and Argentina (agricultural emergency through August)—is without recent precedent in the region's disaster management history. The framework identified this compound configuration 45+ days before the March escalation.

4.7 Region 7: Southeast Asia & Philippines

4.7.1 Predictions Under Assessment

Table 28: Region 7 — Predictions Under March Assessment

ID	Prediction	Prob.	$C(t)$	Window
SEA-1	Early-season WPac tropical cyclogenesis with rapid intensification	70%	0.68	Mar–May
SEA-2	Indonesian urban flooding / landslides (Java, Sulawesi)	75%	0.73	Feb–Mar
SEA-4	Thailand cold surge from Siberian high extension	63%	0.61	Late Feb
—	Severe heatwave, Philippines and mainland SE Asia	82%	0.71	Apr–May
—	Mekong Basin agricultural drought	52%	0.63	Apr–May

4.7.2 Observed Events

Tropical Storm Nuri (March 10–12). Tropical Storm Nuri (03W) formed northeast of Yap, Micronesia on March 10 and was upgraded to tropical storm status on March 11 by JMA—the first March tropical cyclone in the Western Pacific since Typhoon Jelawat in 2018. Nuri peaked at minimal tropical storm intensity with maximum sustained winds of approximately 65 km/h and minimum pressure of 1003 hPa before succumbing to increasing vertical wind shear (50 km/h) and dry-air entrainment. It dissipated on March 12 after an approximately 48-hour lifespan.

Nuri tracked on an unusual northeast trajectory away from the Philippines. PAGASA Chief Administrator Nathaniel Servando assessed a "slim chance to enter PAR." Impacts were limited to scattered rains and thunderstorms across Dinagat Islands, Surigao, and Davao Oriental, with PAGASA warning of possible flash floods.

A second unnamed tropical depression formed to Nuri's east-northeast on March 10 and dissipated March 11 without reaching naming threshold. The January–March total of 3 named storms (Nokaen, Penha, Nuri) matched PAGASA's seasonal outlook of 0–3 for the period. By March 23, JTWC reported no tropical cyclones or invest areas in the basin.

Indonesian Flooding and Landslides (March). Flooding and landslide events continued at significant scale across Indonesia through March, extending the pattern documented in the February report.

Java flooding (March 6–10): At least 7 persons died and 4 remained missing across Bekasi (West Java, 4 deaths), Jember, and Pasuruan (East Java, 3 deaths). More than 53,500 people were affected across Banten, Jakarta, West Java, and East Java. Jakarta flooding on March 7–8 inundated 105 neighborhood units and 19 roads, with water depths reaching 170 cm in Cipete Utara and Pela Mampang (South Jakarta). Bima Regency (West Nusa Tenggara) reported 527 houses and 1 bridge destroyed.

North Kalimantan flooding (March 16–22): Persistent heavy rainfall in Malinau Regency caused major river overflow affecting approximately 43,000 people across 13 villages in 5 sub-districts, with approximately 5,000 houses impacted and 23 education and government facilities damaged. Separately, Nunukan Regency maintained an Emergency Response Status declaration through March 8, with approximately 12,500 people affected by flooding of 1–3 meters depth.

BNPB reported disaster events across 9 provinces during the week of March 16–22. Indonesia's wet season (October–March) was in its late phase, with La Niña weakening toward neutral conditions.

Philippines — Heat Precursors and Dry Season Onset. PAGASA officially declared the end of the northeast monsoon (*amihan*) and the start of the warm dry season on March 23. Heat index readings through March showed escalating temperatures: Coron, Palawan reached 43°C ("Danger" category) and Dumangas, Iloilo reached 42°C on March 2–3. Multiple stations across Mindanao and Visayas recorded heat indices of 37–40°C through mid-March. PAGASA forecast that extreme temperatures of 40.5°C may be reached in Northern Luzon by May.

Thailand — Seasonal Transition. Thailand completed its transition from late cold surges (early March) to pre-monsoon convective regime (mid-March) to full pre-monsoon heat (late March). TMD issued Weather Advisory No. 1 on February 28 warning of summer storms March 3–6, including thunderstorms, hail, and strong winds. Storm damage during March 9–15 affected 519 houses across 8 Northern Region provinces. By late March, Bangkok heat index reached the "dangerous level" (≥42°C threshold), with the Department of Health warning that 2024 saw 30 heat-related deaths in March–April alone.

Vietnam — Mekong Delta Saltwater Intrusion. Peak saltwater intrusion was forecast for end of March 2026 (high tide March 18–22), with the 4 g/L salinity boundary penetrating 44–55 km inland from river mouths—approximately at average level and 5–11 km lower than the 2024–25 season. Ho Chi Minh City issued directives on March 20 activating reservoir operations to push back saltwater. Official assessment: "not severe, lower than or equivalent to average." Approximately 12,266 hectares of winter-spring rice were identified as potentially at risk.

4.7.3 Verification Assessment

Table 29: Region 7 — March Verification Assessment

ID	Prediction	Observed Match	Status	Cumul.
SEA-1	WPac cyclogen. + RI	TS Nuri Mar 10–12; first Mar WPac TC since 2018; 65 km/h; no RI; 2nd unnamed TD	Partial	Partial → Partial
SEA-2	Indonesian flooding	Java: 7 dead, 53.5k affected; N. Kalimantan: 43k affected; Jakarta 170 cm; multi-island	Confirmed	Conf. → Conf.
SEA-4	Thailand cold surge	Confirmed Feb (Feb 24–28); Mar: pre-monsoon transition; no add'l surges	Partial	Partial → Partial
—	SE Asia heatwave	PH dry season declared Mar 23; heat indices 42–43°C; Bangkok "dangerous"; window Apr–May	Developing	New
—	Mekong drought	Saltwater intrusion avg levels; 12,266 ha rice at risk; "not severe"; window Apr–May	Developing	New

Assessment Notes. Prediction SEA-1 remains classified as Partial. The framework predicted early-season Western Pacific tropical cyclogenesis with rapid intensification potential. Tropical Storm Nuri confirmed the cyclogenesis component—it was the first March tropical cyclone in the basin since 2018, and the January–March total of 3 named storms confirms the anomalously active state of the warm pool (+2.0°C above climate average) that the framework identified. However, Nuri peaked at minimal tropical storm intensity (65 km/h) and did not exhibit rapid intensification. The prediction's coherence value of 0.68 (Moderate, the lowest in the regional set) appropriately reflected the uncertainty in whether the thermodynamic potential would translate to individual storm intensification versus distributed low-intensity cyclogenesis.

The rapid intensification component of this prediction has a window extending through May 2026. As ENSO transitions toward El Niño and Western Pacific SSTs remain anomalously warm, the conditions for rapid intensification will become more favorable through the April–May window. The Partial classification may be revisited if a qualifying RI event occurs within the specified window.

Prediction SEA-2 continues to confirm at a level consistent with or exceeding the February assessment. The February report documented 80 fatalities from the West Bandung landslide alone. March added 7 confirmed dead in Java, 43,000 affected in North Kalimantan, and simultaneous multi-island flooding across 9 provinces. The cumulative February–March toll of 87+ fatalities and hundreds of thousands affected across Java, Kalimantan, Bali, and Nusa Tenggara represents one of the most active wet seasons in recent Indonesian history.

The Philippines heatwave and Mekong drought predictions, with April–May windows, show early precursor signals consistent with the framework's assessed trajectory. PAGASA's dry season declaration on March 23 and heat indices already reaching "Danger" category suggest the framework's 82% probability heatwave prediction is on track.

4.7.4 Economic and Human Impact

Table 30: Region 7 — March Impact Summary

Metric	Observed
Fatalities	7+ (Indonesia Java flooding); additional Kalimantan figures under compilation
Persons affected	96,500+ (Java 53,500; N. Kalimantan 43,000; additional provinces pending)
Infrastructure	527 houses + 1 bridge destroyed (Bima); 5,000 houses impacted (Malinau); Jakarta: 105 neighborhood units inundated
Cyclone activity	3 named storms Jan–Mar (matching PAGASA seasonal outlook); first Mar TC since 2018
Heat precursors	Philippines: heat indices 42–43°C; Bangkok: "dangerous level"; Thailand: 519 houses damaged by pre-monsoon storms
Cumulative Feb–Mar	Indonesia: 87+ fatalities, hundreds of thousands affected across 4+ major islands

5 Compound Risk Verification — March Update

The three compound risk scenarios issued on February 5, 2026 were assessed in the February report. This section updates each with March observational data.

5.1 Compound Risk 1: California Flush — March Evolution

Table 31: Compound Risk 1 — California Flush: Cumulative Assessment

Component	Cumulative Feb–Mar Observed	Status
Multi-wave AR sequence	Three waves Feb 10–25; window exact; no significant Mar ARs	Confirmed
Burn scar debris flows	Palisades/Eaton/Hurst activated Feb; 1 fatality	Confirmed
Elevated freezing levels → rain-on-snow	Initial 6,500 ft confirmed; dropped to 2,500 ft wave 3; Castle Peak avalanche (9 dead)	Partial
Sierra snowpack stress	Feb: 68% of normal; Mar 24: **34% of normal**; melting at ∼1%/day under heat dome	Confirmed (Exceeded)
Compound assessment		**Confirmed**

The California compound scenario evolved significantly between February and March. The AR sequence and debris flow components were confirmed in February and did not recur in March (the pattern shifted to the persistent ridge/heat dome). The snowpack component, however, escalated dramatically: the Sierra Nevada declined from 68% of normal at the end of February to 34% by March 24, with DWR warning of the potential for the second-lowest April 1 snowpack in recorded history.

The March heat dome represents a hazard-mode switch within the same compound scenario. The framework predicted "the California Flush"—a sequence in which AR precipitation, burn scar activation, and snowpack stress converge. The observed outcome delivered this convergence in February (AR + debris flows + rain-on-snow/avalanche risk) and then shifted to a different mode of snowpack destruction in March (extreme heat + accelerated melt). The compound risk materialized across both months through different mechanisms operating on the same underlying vulnerability.

This observation reinforces the February report's finding regarding hazard-mode switching: the framework correctly identified the forcing (persistent ridge + anomalous ocean heat) and the vulnerability (depleted snowpack, recent burn scars) but did not anticipate the thermodynamic phase transition between precipitation-driven and heat-driven snowpack loss. The March heat dome's acceleration of snowpack melt represents a more severe long-term water supply outcome than the rain-on-snow flooding the framework specified, as melted snow exits the system entirely rather than producing potentially useful (if hazardous) runoff.

5.2 Compound Risk 2: Eurasian Freeze-Thaw Whiplash — March Update

Table 32: Compound Risk 2 — Eurasian Freeze-Thaw Whiplash: Cumulative Assessment

Component	**Cumulative Feb–Mar Observed**	**Status**
SSW-driven cold, W. Europe	Feb: cold confined to Scandinavia; late Mar: "Great Reversal" directing cold into W. Europe via Greenland blocking	Partial → Developing
China cold wave sequence	5+ events confirmed (4 Feb + ≥1 Mar)	Confirmed
Dust storm during thaw transition	Feb 21–23: AQI 500, 130+ cities; Mar 13: Xinjiang event	Confirmed
Spring Festival travel disruption	9.5B trips; dust storm hit return wave	Confirmed
Compound assessment		**Partial**

The Eurasian compound scenario's Western European cold component has evolved in an interesting direction. The February report classified it as Partial because the SSW-driven cold was routed to North America rather than Western Europe. The March SSW event, combined with developing Greenland blocking, is now directing an Arctic plunge into Western Europe in late March—the "Great March Reversal" described by Severe Weather Europe. This late-March cold event, with forecast temperature drops of 15–20°C and frost risk to budding vineyards, represents a delayed manifestation of the predicted SSW → blocking → cold air displacement mechanism. The "thaw" component (rapid warming after cold) has not yet materialized; if the late-March cold is followed by a rapid warm-up in April, the full freeze-thaw whiplash cycle could complete within the remaining prediction window.

The compound assessment remains Partial because the Western European cold component has not yet produced the severe surface impacts (ice storm, infrastructure disruption) specified in the high-coherence scenario. However, the mechanism chain is now verifying with a temporal lag of approximately 4–6 weeks relative to the original prediction window. The v1.3 development note from the February report—incorporating vortex lobe geometry as an explicit routing parameter—is reinforced by the March data: the February SSW routed cold westward to North America, while the March SSW is routing cold southward into Western Europe. Both are valid SSW responses; the framework's current coherence tensor does not discriminate between them with sufficient geographic precision.

5.3 Compound Risk 3: South American Dipole — Sustained

Table 33: Compound Risk 3 — South American Dipole: Cumulative Assessment

Component	Cumulative Feb–Mar Observed	Status
SE Brazil SACZ flooding (SA-1)	73 dead (Feb); continued Mar flooding in SP/RJ; SACZ mechanism exact	Confirmed
Coastal El Niño, Peru (SA-2)	92 dead; 1,177 districts (62%); Ecuador national emergency; Niño 1+2 at +1.5°C	Confirmed
Pampas drought (SA-3)	Formal Agricultural Emergency declared Mar 1–Aug 31 despite partial rainfall relief	Confirmed
Compound assessment — all three legs sustained for two consecutive months		**Confirmed**

The South American Dipole remains the framework's strongest compound risk verification. All three legs have been simultaneously active for two consecutive months without interruption. The framework's prediction document stated that these events are "geometrically linked through the South American monsoon system's moisture transport architecture"—a coupled mode in which SACZ stalling over southeastern Brazil diverts moisture from the Pampas while Walker circulation anomalies reinforce coastal warming along Peru's littoral.

March data strengthens this assessment. The SACZ continued to produce heavy rainfall across southeastern Brazil while the Pampas experienced partial relief only through transient frontal rainfall—not through a shift in the monsoon architecture. The Coastal El Niño intensified as Niño 1+2 warmed from near-zero to +1.5°C. Argentina declared formal agricultural emergency even as crop conditions improved, indicating that the structural drought stress identified by the framework required institutional intervention regardless of short-term precipitation.

The two-month sustained co-occurrence of all three legs moves this result beyond what single-month verification can establish. Independence-based models would treat each month as a separate trial; the framework's geometric coupling argument predicts sustained co-occurrence as long as the underlying monsoon architecture maintains its configuration. The observed persistence is consistent with the coupled-mode hypothesis.

6 Aggregate Impact Summary

Table 34: March 2026 — Aggregate Impact by Region

Region	Fatal.	Displacement / Affected	Est. Economic Loss	Lead
1. Australia	0	500+ evacuated (Numbulwar); Daly River community; Katherine	Pilbara industrial risk pending; Riverland crop losses	34+ d
2. China	TBD	Data limited	Data limited	—
3. United States	15+	Tens of thousands (tornadoes, blizzard, wildfire)	Multi-billion USD (wildfire, tornado, snowpack/water)	30+ d
4. W. Europe	5+	Groundwater flooding persistent; avalanche communities isolated	>€4B cumulative (Iberian); PV efficiency losses	8–17 d
5. Middle East	10+	106+ displaced (Oman); Mar 25–31 impacts pending	Pending; Baghdad flood risk acute	49+ d
6. S. America	106+	29,000+ displaced (Peru + Ecuador); 10,000 (Brazil)	$84M+ Peru roads/agriculture; Ecuador crop/livestock losses	45+ d
7. SE Asia	7+	96,500+ affected (Indonesia)	$25M+ (Philippines cumulative); Indonesia pending	3–18 d
March Total	**>143**	**>136,000**	**Est. $8–12B+**	—
Feb–Mar Cumulative	**>454**	**>936,000**	**Est. $20–25B+**	—

Economic loss estimates reflect publicly reported figures as of March 25, 2026. Several institutional assessments—notably for the Western U.S. heat dome and wildfire complex, Tropical Cyclone Narelle's WA landfall (in progress), and the Middle East March 25–31 storm system—had not been finalized at the time of publication. The $8–12 billion March estimate and $20–25 billion cumulative estimate are assessed as conservative; final institutional tallies are expected to revise these figures upward.

The fatality distribution continues the pattern identified in the February report: developing-nation contexts (South America: 106+; Middle East: 10+; Southeast Asia: 7+) account for the majority of the global total despite representing three of seven assessed regions. Peru alone accounts for more March fatalities (92) than the United States (15+), Western Europe (5+), and Australia (0) combined—a disparity that reflects differences in exposure, infrastructure resilience, and emergency response capacity rather than meteorological severity.

7 Framework Performance Assessment

7.1 March Standalone Verification Statistics

Table 35: March 2026 Standalone Verification Statistics

Classification	Count	%	Interpretation
Confirmed	8	53.3%	Event matched window, region, and mechanism
Confirmed (Exceeded)	2	13.3%	Confirmed with observed magnitude substantially exceeding thresho
Partial	3	20.0%	Mechanism confirmed; geographic or magnitude deviation
Developing	1	6.7%	Window open; precursor signals confirmed
Not Verified	1	6.7%	No observed match within specified window
Total	**15**	**100%**	

7.2 Cumulative Reclassification Statistics

Table 36: Cumulative February–March Reclassification (Original 28 Predictions)

Classification	February (Published)	Cumulative	Change
Confirmed	16 (57.1%)	18 (64.3%)	+2
Confirmed (Early)	4 (14.3%)	1 (3.6%)	−3 (reclassified to Confirmed)
Confirmed (Exceeded)	—	2 (7.1%)	+2 (new classification)
Total confirmed class	**20 (71.4%)**	**21 (75.0%)**	**+1**
Partial	5 (17.9%)	4 (14.3%)	−1 (AU-1 upgraded)
Developing	3 (10.7%)	3 (10.7%)	0
Not Verified	0 (0.0%)	0 (0.0%)	0

The cumulative confirmed rate rose from 71.4% to 75.0% on the same 28-prediction set. The increase reflects one Partial → Confirmed upgrade (AU-1: Queensland tropical cyclone) and three Confirmed (Early) → Confirmed reclassifications as events that arrived ahead of their specified windows were joined by within-window events (CN-3, ME-2, SA-2). Two predictions received the new Confirmed (Exceeded) designation (US-5, US-7) reflecting observed outcomes substantially surpassing predicted thresholds.

No prediction confirmed in February was contradicted by March observations. Every confirmed prediction either maintained its status or escalated in severity. This persistence pattern is consistent with the framework's core design premise: geometric constraints, once identified, produce sustained atmospheric modes rather than isolated events.

7.3 Coherence Value Calibration — Two-Month Update

Table 37: Mean $C(t)$ by Cumulative Verification Outcome

Outcome	Mean $C(t)$	Mean Probability
Confirmed + Confirmed (Exceeded)	0.74	75.8%
Confirmed (Early)	0.64	66.0%
Partial	0.65	66.5%
Developing	0.73	76.3%
Not Verified	0.76	78.0%

The two-month calibration data reveals a finding that warrants attention: the single Not Verified prediction (AU-2, SA/VIC heatwave) had the highest $C(t)$ value (0.76) and highest probability (78%) of any non-confirmed prediction. Combined with the February finding that the lowest-probability European prediction (EU-3, 65%) produced the highest-severity outcome, these results suggest the coherence tensor may exhibit different calibration characteristics for different hazard classes. Specifically:

- **Synoptic-scale storm systems and cyclogenesis:** Higher $C(t)$ correlates with higher confirmation rates, consistent with design intent.
- **Persistent ridge/blocking dynamics:** The relationship is less reliable. Both the Not Verified heatwave (high $C(t)$, ridge broke down) and the Exceeded snowpack deficit (low $C(t)$, ridge persisted far beyond expectations) involved blocking pattern persistence.
- **Persistent weak layers (avalanche, snowpack):** Low assigned $C(t)$ has consistently correlated with extreme outcomes, suggesting systematic underweighting.

This hazard-class-dependent calibration signal is the most actionable finding from the two-month verification series for v1.3 development.

7.4 Lead Time Assessment — March

Framework predictions were issued on February 5, 2026. March events verified at lead times of 30–49 days:

- **TC Narelle QLD landfall (March 20):** 43 days lead time from issuance to formation; 34+ days to the BOM prediction window for Coral Sea cyclogenesis.
- **EF5 Indiana tornado (March 10):** 33 days. No operational center identified EF5 potential in early March for the Midwest at that range.
- **Peru emergency escalation (1,177 districts, March 20):** 43 days from issuance to the district count exceeding 1,000.
- **Middle East storm system (March 25):** 48 days. Operational NWP models resolved the synoptic pattern approximately 5–7 days before onset.

The framework's lead-time advantage relative to operational NWP widened in March compared to February (where lead times ranged from 3–23 days). This is expected: as events occur further from the issuance date, the gap between geometric constraint identification (weeks to months) and synoptic-scale NWP resolution (days) increases. The framework does not compete

with NWP on spatial or temporal precision; its contribution is identifying elevated-probability windows and compound risk configurations at temporal scales beyond NWP's deterministic horizon.

7.5 Systematic Observations — Two-Month Patterns

Four systematic patterns have emerged from the February–March verification:

1. Escalation rather than regression. No confirmed prediction regressed to unconfirmed status in March. Five predictions escalated in severity (AU-1, US-5, US-7, SA-2, ME-2). This is the pattern expected from a framework that identifies persistent atmospheric modes: if the geometric constraints remain in place, subsequent events should be of comparable or greater magnitude as the mode matures.

2. Conservative temporal calibration confirmed. The February report identified an early-arrival bias (4 predictions arriving 1–4 weeks ahead of specified windows). March data resolved this: three of four "Early" predictions were joined by within-window events, confirming the predictions at both temporal positions. The framework's temporal windows were not systematically early—they were conservatively calibrated, and the atmospheric modes produced events across the full window duration.

3. Hazard-mode switching within compound scenarios. Both the California Flush (AR/debris flow → heat dome/melt) and the Eurasian Freeze-Thaw (SSW → N. America cold in Feb → W. Europe cold in Mar) demonstrated that persistent atmospheric modes can produce different hazard types across successive months. The framework identified the forcing mode correctly in both cases but did not anticipate the specific hazard transition. V1.3 development should incorporate precipitation phase sensitivity and vortex lobe routing as explicit parameters.

4. Compound risk sustained co-occurrence. The South American Dipole maintained simultaneous three-leg activation for two consecutive months. The probability of this under independence assumptions (∼39% for a single month) decreases to ∼15% for two consecutive months. The observed sustained co-occurrence provides additional evidence for the geometric coupling hypothesis over the independence assumption.

8 On Misses — What the Heat Dome Reveals About the System's Architecture

The Western U.S. heat dome of March 2026 is the most significant event within the framework's assessment domain that was not predicted. It warrants more than a passing mention in the honest framing section. It warrants analysis—because the nature of the miss is more informative than most of the hits.

The Auburn Coherence Framework v1.2 is a geometric constraint system. It identifies persistent structures in atmospheric flow—ridges, troughs, blocking patterns, monsoon architectures—that have minimized their internal dissipation pathways and are therefore geometrically protected from decay. The coherence value $C(t)$ measures the degree of this geometric alignment. When $C(t)$ is high, the structure persists. When it decays, the structure breaks down. That is the entire engine.

The framework *did* identify the persistent North Pacific Ridge. The February 5 prediction document described it as a "high-coherence blocking configuration" that had "deflected the storm track north of California for weeks." It correctly predicted the ridge would suppress precipitation and degrade Western snowpack. It assigned a snowpack deficit prediction (US-5) that verified at record-breaking levels. The geometry was correct.

What the framework did not resolve was the *evolution within the persistent structure*—how the ridge amplified during March, interacted with the SSW-driven hemispheric reconfiguration,

and produced a 500-year return period heat event rather than continued warm-and-dry conditions. The framework saw the shape. It did not see the shape intensify into an extreme state.

This is not a random failure. It is a *specific architectural boundary*, and it maps onto a specific body of mathematics.

8.1 The Geometric-Only Boundary

The v1.2 framework was built on geometric constraints derived from Navier-Stokes dynamics. But it was built *before* the author's subsequent work on the Angular Cancellation Lemma (ACL) and the closure of the two-enstrophy gap—mathematical results that describe how energy cascades within and between persistent structures. The coherence tensor identifies when a structure is geometrically stable. What it does not yet capture is the energy accumulation dynamics within a stable structure—how a ridge that is geometrically protected from breakdown can simultaneously be loading energy that eventually manifests as an extreme exceedance of its normal behavioral envelope.

That is precisely what enstrophy dynamics describe. A ridge that is "coherent" in the v1.2 sense (geometrically persistent, low internal dissipation) can simultaneously be building enstrophy—concentrating vorticity in a configuration that transforms a "warm and dry" atmospheric mode into a historic, 14-state-record-shattering heat dome. The v1.2 system does not distinguish between a coherent structure at equilibrium and a coherent structure accumulating energy toward an extreme state. The mathematical tools to make that distinction now exist in the author's other published work. They were not available when v1.2 was designed.

8.2 Three Findings from the Miss

The heat dome miss reveals three things.

First, the geometric layer alone—persistent structure identification and decay timeline estimation—captures a larger fraction of atmospheric predictability than the existing literature would suggest. A 75% confirmed rate over two months, with zero contradictions, on a system the author considers fundamentally incomplete, indicates that geometric constraints are doing more explanatory work than conventional approaches typically attribute to them.

Second, the boundary of the geometric-only approach is identifiable and specific: it lies at the transition between "this structure will persist" and "this structure will persist *and intensify toward an extreme state*." That transition requires energy cascade dynamics that the coherence tensor does not model.

Third, the mathematical apparatus to extend across that boundary—the ACL, the enstrophy closure, the energy cascade formalism—exists in the author's published work but has not been integrated into the atmospheric prediction framework. The miss is a roadmap.

8.3 On the Question of v1.3

The author has no current intention of building it.

The Auburn Coherence Framework prediction series covers February through May 2026 and will not be extended. The framework was designed as a demonstration—a proof that geometric constraint identification produces operationally useful atmospheric predictions at temporal scales beyond numerical weather prediction's deterministic horizon. The February and March verification reports establish that proof at 75% confirmed with zero contradictions across 28 assessed predictions.

The integration of ACL and enstrophy dynamics into the coherence tensor—which would address the heat dome class of misses—is a substantial research program. It is available to whoever carries this work forward: an institutional partner, a research group with the computational

infrastructure to operationalize what the author has demonstrated in theoretical form, or a commercial entity prepared to license the underlying mathematics.

The author's role was to prove the geometric layer works. That proof is now timestamped, published, and verified against two months of global observational data. What comes next is someone else's problem.

Honest Framing

This section is included because the Auburn Coherence Framework's credibility depends on transparent analysis of its failures, not just its successes. A framework that publishes only its hits is marketing. A framework that analyzes its misses, identifies the specific architectural boundary they reveal, and maps that boundary onto existing mathematics—without promising to fix it—is research.

9 Honest Framing

Honest Framing

The Auburn Coherence Framework provides probabilistic risk identification based on geometric constraint analysis of atmospheric dynamics. It does not provide deterministic forecasts, and a verification report documenting confirmed predictions does not establish predictive infallibility.

The following limitations apply to this report and to the framework's methodology:

1. The heat dome miss matters. The most impactful March weather event in the United States—a heat dome that shattered March temperature records in 14 states, was assessed by World Weather Attribution as requiring anthropogenic climate change to be physically possible, and triggered the largest wildfire in Nebraska history—was not predicted by the framework. This event occurred within the framework's geographic and temporal domain. The mechanistic linkage to predicted snowpack and drought dynamics does not substitute for explicit prediction. A framework that captures 75% of assessed predictions but misses a 500-year return period event in its core assessment region has demonstrated both capability and boundaries.

2. Base-rate problem persists. March is a climatologically transitional month with elevated severe weather potential across multiple regions. Some fraction of confirmed predictions would have been issued by any competent meteorological assessment. The framework's value is measured not by predicting "storms in March" but by specifying mechanisms, windows, compound interactions, and escalation trajectories that conventional assessments did not link.

3. Small sample size. Forty-three predictions across two months do not constitute a statistically robust calibration dataset. The 75% cumulative confirmation rate cannot be treated as a confidence interval on future performance. Multi-year verification across hundreds of predictions is required before probability assignments can be validated against observed frequencies.

4. Data gaps are asymmetric. The China and Middle East assessments operate under structural data limitations (language barriers, conflict-degraded observation capacity) that do not affect other regions. The framework's apparent performance in data-rich regions (US, Australia, Europe) may not be representative of its performance in data-poor regions. The Developing classifications for CN-5 and ME-1 could resolve to either Confirmed or Not Verified when data becomes available; the direction of resolution is unknown.

5. Confirmation bias in sequential reports. A second verification report inherently builds on the framing of the first. The February report's confirmed predictions create an expectation of continued verification that may bias the March assessment toward confirmation. This report mitigates this risk by (a) applying identical classification criteria, (b) documenting the Not Verified prediction (AU-2) and the unpredicted heat dome with equal prominence, and (c) maintaining the honest framing section as a structural requirement rather than an afterthought.

6. The framework did not predict the Iran conflict. Several March outcomes—degraded Middle Eastern observation capacity, the Tehran "black rain" event, amplified humanitarian vulnerability to weather hazards in conflict zones—resulted from geopolitical events entirely outside the framework's domain. No atmospheric prediction framework can anticipate military conflicts. The framework's Middle East predictions were issued assuming baseline observation and emergency response capacity; the conflict has fundamentally altered the risk landscape in ways the framework cannot capture.

This framework is analogous to financial auditing: it certifies process compliance and risk identification without guaranteeing future outcomes. The February–March verification demonstrates that the geometric constraint methodology identified real atmospheric modes with operationally useful lead times across two consecutive months. It does not guarantee that April–May predictions will achieve comparable verification rates.

PUBLIC RELEASE – NON-COMMERCIAL

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Appendix A: Full Prediction-by-Prediction Verification Matrix — March 2026

The following table provides the complete March verification record for predictions assessed in this report. Each entry includes the prediction identifier, region, prediction summary, March observed outcome, March status, and cumulative February–March status.

Table 38: Complete March Verification Matrix

ID	Reg.	Prediction	Prob.	$C(t)$	March Observed Outcome	Mar Status	Cumul.
AU-1	1	Cat 3+ cyclone, QLD	72%	0.71	TC Narelle Cat 5 peak; Cat 4 QLD landfall Mar 20; triple landfall QLD/NT/WA	Conf. (Exc.)	Conf.
AU-2	1	Heatwave SA/VIC ≥45°C	78%	0.76	No heatwave; Riverland flooding instead	Not Verif.	Not Verif.
AU-3	1	Bushfire continuation	80%	0.82	Jan season confirmed; Mar suppressed by rain	Confirmed	Conf.
AU-4	1	Murray-Darling deficit	77%	0.74	Storage 48% (19% below prior yr); partial rain relief; dry outlook	Confirmed	Conf.
AU-5	1	Delayed Autumn Break	82%	0.79	BOM: below-avg rain 60–80% Apr–Jun	Developing	Devel.
CN-2	2	Cold wave ≥3	74%	0.72	≥5 total events (4 Feb + ≥1 Mar)	Confirmed	Conf.
CN-3	2	Dust storm Beijing/NE	75%	0.73	Mar 13 Xinjiang; did not reach Beijing/Korea	Partial	Partial
CN-5	2	SE China drought	77%	0.75	NMC advisory; no Mar quant. data in English	Developing	Devel.
CN-6	2	Tibetan snowfall	62%	0.60	Confirmed Feb; no add'l Mar data	Confirmed	Conf.
US-1	3	CA AR sequence	81%	0.80	Feb confirmed; Mar shifted to heat dome	Confirmed	Conf.
US-5	3	Snowpack <75%	67%	0.65	SNOTEL-era record lows; CA 34%; CO lowest since 1981	Conf. (Exc.)	Conf. (Exc.)
US-7	3	Severe weather / tornadic	64%	0.62	∼150 tornadoes (∼2× avg); EF5 IN; EF3 earliest MI	Conf. (Exc.)	Conf. (Exc.)
EU-1	4	Iberian flooding	84%	0.83	Storm sequence continued; 19 named storms	Confirmed	Conf.
EU-2	4	SSW cold (Beast from East)	68%	0.66	Late-Mar "Great Reversal"; cold now routing to W. Europe	Partial	Partial

Continued on next page

Table 38 — continued

ID	Reg.	Prediction	Prob.	$C(t)$	March Observed Outcome	Mar Status	Cumul.
EU-3	4	Alpine avalanche 4–5	65%	0.63	127 fatalities (7–8 yr high); Level 4 CH; Val Ridanna 2 dead	Confirmed	Conf.
EU-4	4	UK/Rhine flooding	72%	0.70	UK groundwater flooding persistent; Rhine: no event	Partial	Partial
EU-5	4	Scandi. blocking + NAO	80%	0.79	Winter confirmed; Mar shift; late-Mar revert to blocking	Confirmed	Conf.
ME-1	5	Dust sequence	88%	0.86	No confirmed major events; observation degraded	Developing	Devel.
ME-2	5	Flash flooding, jet superpos.	62%	0.60	Oman 10 dead; Mar 25–31: 100–500 mm forecast Iraq	Confirmed	Conf.
ME-3	5	Fertile Crescent drought	85%	0.84	Worst since 1933; reservoirs at construction-era lows	Confirmed	Conf.
SA-1	6	SE Brazil SACZ	79%	0.78	73 dead final; Mar flooding continued SP/RJ	Confirmed	Conf.
SA-2	6	Coastal El Niño, Peru/Ec.	66%	0.64	92 dead; 1,177 districts; Ecuador nat'l emergency	Conf. (Exc.)	Conf.
SA-3	6	Pampas drought	74%	0.72	Partial relief; formal Agric. Emergency declared Mar 1–Aug 31	Confirmed	Conf.
SEA-1	7	WPac cyclogenesis + RI	70%	0.68	TS Nuri Mar 10–12; 65 km/h; no RI	Partial	Partial
SEA-2	7	Indonesian flooding	75%	0.73	Java 7 dead, 53.5k affected; N. Kalimantan 43k affected	Confirmed	Conf.
SEA-4	7	Thailand cold surge	63%	0.61	Confirmed Feb; Mar: seasonal transition	Partial	Partial

Appendix B: Post-Publication Update (April 2, 2026)

This appendix documents observational data received after the report's data collection cutoff of March 25, 2026. No prediction classifications are changed by this update. The purpose is to complete the observational record for events that were in progress or forecast-only at the time of publication.

AU-1: Tropical Cyclone Narelle — Fourth Landfall Confirmed

Narelle made a fourth and final Australian landfall at approximately 09:30 AWST on March 27, crossing the Gascoyne coast just south of Coral Bay, Western Australia, as a Category 3 system. At landfall, sustained winds were approximately 140 km/h with gusts to 195 km/h. Learmonth Airport recorded wind speeds of approximately 200 km/h and rainfall exceeding 300 mm. Exmouth airport was destroyed. Exmouth sustained severe damage, with approximately 1,400 properties without power as of March 29. LNG operations across the Pilbara were shut down. A dangerous storm tide developed along the coast between Onslow and Denham. The system generated a dramatic dust storm across arid Western Australia, producing blood-red skies in Shark Bay and Denham as iron oxide-rich soil was lofted into the atmosphere. No fatalities or serious injuries were reported from the WA landfall.

AccuWeather confirmed Narelle as a four-landfall event (QLD March 20, NT March 21–22, Kimberley March 23, Gascoyne March 27), exceeding the historical comparison to Cyclone Ingrid (2005) and Cyclone Steve (2000), which each made three mainland landfalls. NASA Earthdata documented the full track from March 19–27 using VIIRS imagery. Zoom Earth recorded peak winds of 230 km/h during Narelle's lifetime.

The fourth landfall does not change the Confirmed (Exceeded) classification assigned in the main report. It does, however, elevate the historical significance of the event: Narelle is now the only tropical cyclone in the modern Australian record to make four distinct mainland landfalls.

ME-2: March 25–31 Storm System — Observational Confirmation

The storm system described as "active/developing" in the main report delivered significant impacts across the UAE and broader Gulf region between March 25 and 27. The Watchers confirmed widespread flooding across Dubai, Abu Dhabi, and the Northern Emirates. Jebel Yanas in northern UAE recorded 244 mm—more than double the typical annual rainfall range of 60–100 mm for the region. Multiple UAE stations exceeded 100 mm during the event. Sharjah suffered the worst urban flooding, described by The National as a repeat of the April 2024 deluge. Roads were inundated, vehicles stranded, and flight operations disrupted at both Dubai International and Zayed International airports. Municipal teams conducted drainage operations through the weekend.

The storm's impacts were compounded by the ongoing Iran-UAE military conflict. As of March 31, the UAE had intercepted 433 ballistic missiles, 1,977 drones, and 19 cruise missiles fired from Iran since February 28, according to the UAE Ministry of Defence. The convergence of extreme weather and active military operations across the same geographic area created compound emergency response demands that neither hazard alone would have produced—precisely the scenario the Honest Framing section of this report identified as beyond the framework's predictive domain.

The observational confirmation of the March 25–31 storm system does not change the Confirmed classification for ME-2, which was established on the basis of the earlier Oman flooding (10 fatalities, March 20–24). The late-March data does, however, complete the verification: the framework predicted flash flooding from Red Sea Trough/jet superposition in the March–April window, and the observed atmospheric configuration—CAPE values of 4,000–5,000 J/kg, subtropical jet interaction, surface cyclone near Baghdad—matched the predicted mechanism

with high fidelity. The 244 mm at Jebel Yanas is comparable to the April 2024 benchmark event (164 mm at Dubai International) that the framework used as its primary analog.

Summary of Post-Publication Updates

Honest Framing

No prediction classifications are revised by this appendix. The confirmed rate for the original 28 predictions remains 75.0% (21/28). The March standalone rate remains 66.7% (10/15). The two updates documented above replace forecast-dependent evidence with observational confirmation, strengthening the empirical basis of existing classifications without altering them.

The complete March 2026 observational record now includes data through March 31 for all seven regions. Three data gaps identified in the main report persist: Chinese-language NMC drought monitoring for southeastern China, Middle Eastern dust storm observation capacity degraded by conflict, and April 1 snowpack measurements (which were released after the report cutoff and, based on preliminary reporting, confirmed record or near-record lows across most Western U.S. basins—consistent with the Confirmed (Exceeded) classification for US-5).

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Auburn Coherence Framework v1.2

"Standardize the protocol, let the content evolve."